

Picard Iteration for the Characteristic Initial Value Problem in Einstein Equations

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8 September 2026

Abstract

We present an iteration algorithm for vacuum and Einstein scalar-field equations in double-null gauge, which transform the non-linear PDE into systems of ODE. The numerical realization combines characteristic constraint solves, LGL spectral elements, pole-free spherical operators, Galerkin projection, and independent first-order residual and consistency checks.

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1 Introduction

The characteristic initial value problem is a natural formulation of the Einstein equations when radiation, null focusing, and the formation of trapped surfaces are central. Instead

of prescribing data on a spacelike hypersurface, one prescribes compatible data on two intersecting null hypersurfaces and evolves their future domain of dependence. Local well-posedness for this problem is known under suitable regularity assumptions [19, 16, 5, 14], but its direct numerical realization becomes substantially more difficult once spherical symmetry is removed. The use of null hypersurfaces as an evolution framework goes back to the Bondi–Sachs description of gravitational radiation and the early analysis of characteristic data [4, 20]; its development as a numerical method is surveyed in [21].

Characteristic numerical relativity has progressed from double-null vacuum evolutions with two Killing symmetries [8] and spherically symmetric scalar collapse in null coordinates [10, 13] to axisymmetric vacuum evolution on outgoing null cones [11] and generic angular data for accurate Cauchy–characteristic waveform extraction [18]. Numerical relativity has also long evolved nonspherical spacetimes in spacelike foliations. Fully three-dimensional Cauchy calculations have studied scalar-field collapse without symmetry assumptions [9], and axisymmetric pseudospectral calculations have reached strongly aspherical regimes in which the center of collapse bifurcates [17]. The first null-coordinate simulations of nonspherical regular data collapsing to a black hole were obtained in twist-free axisymmetry for the massless scalar field [12]; that work describes the accessible data as moderately nonspherical and identifies larger deviations from spherical symmetry, as well as vacuum collapse, as important challenges. Thus the gap addressed here is not an absence of nonspherical numerical solutions in general. It is the relative scarcity of direct characteristic evolutions with strongly nonspherical, freely prescribed initial geometry in a fully angular double-null gauge.

Let (u, v, θ^A) be double-null coordinates. We write the spacetime metric as

$$\mathbf{g} = -2\Omega^2(\mathrm{d}u \otimes \mathrm{d}v + \mathrm{d}v \otimes \mathrm{d}u) + g_{AB}(\mathrm{d}\theta^A - b^A \mathrm{d}u) \otimes (\mathrm{d}\theta^B - b^B \mathrm{d}u). \quad (1.1)$$

The metric is therefore represented by the section metric g_{AB} , the lapse Ω , and the angular shift b . This formulation gives considerable freedom to choose the geometry on the two initial null hypersurfaces, subject to the characteristic constraints and corner compatibility. In particular, the nonspherical part of the initial metric variation can be prescribed directly rather than produced only indirectly from a matter perturbation or a spacelike constraint solve. The fourth experiment gives a quantitatively strong example: on its outgoing initial hypersurface, where $\Omega = 1$, the prescribed data satisfy

$$\|\mathrm{tf} \mathcal{L}_{\partial_v} g(-1, 0.005)\|_{L^2(S)} = 7.50960, \quad \|\mathrm{tf} \mathcal{L}_{\partial_v} g(-1, 0.005)\|_{L^\infty(S)} = 2.53468. \quad (1.2)$$

The perturbation is consequently a substantial geometric departure from spherical symmetry, rather than a nonspherical profile whose effect on the initial metric is numerically negligible.

This paper develops a first-order Picard iteration for both the Einstein vacuum equations

$$\mathrm{Ric}(\mathbf{g}) = 0 \quad (1.3)$$

and the Einstein–massless-scalar system

$$\mathrm{Ric}(\mathbf{g}) = \mathrm{d}\phi \otimes \mathrm{d}\phi, \quad \square_{\mathbf{g}}\phi = 0, \quad (1.4)$$

in the gauge (1.1). The iteration is organized by the geometry of the characteristic problem. Data that are free, data fixed by the null constraints, and quantities transported in the two characteristic directions play different roles in each sweep. The resulting construction is therefore not obtained by treating all coordinate components as an undifferentiated system of equations: its update order follows the causal and geometric dependency structure of the double-null equations.

The iteration architecture is adapted from the approximate-spacetime construction introduced by An and Wu [2]. Both constructions use a triangular hierarchy of double-null transport equations, with nonlinear coefficients evaluated at the preceding iterate. The present work modifies that analytic construction into an executable numerical Picard map: it uses numerically convenient weighted variables and equivalent transport identities, imposes both characteristic traces discretely, and incorporates projection, relaxation, and residual auditing. Thus the underlying geometric organization is largely the same, while some update equations and terms are reformulated for numerical computation.

Two difficulties are particularly important. The first is to obtain an executable form of the equations. Analytic arguments can often group lower-order terms schematically or pass between conventionally equivalent forms, whereas numerical evolution requires every displayed sign and numerical factor to be fixed consistently. We address this by checking the formulas against identities derived from the four-metric, by testing exact solutions, and by evaluating first-order residuals using fresh derivatives of the stored weighted fields rather than the construction sources of a Picard sweep. We check the metric–connection consistency separately and avoid second null derivatives at the rough characteristic endpoint. The second difficulty is the iteration itself. Its design requires identifying a closed hierarchy that respects the characteristic constraints, transports information in the correct direction, preserves the two initial traces, and remains meaningful as the null hypersurfaces focus. This construction comes from the geometric structure of the equations rather than from a generic fixed-point template.

The numerical implementation combines Legendre–Gauss–Lobatto spectral elements in the two null coordinates with pole-free spherical differentiation, angular Galerkin projection, characteristic constraint solves, and independent overgrid audits. Extensive AI-assisted software development made it practical to implement, refactor, and test this large coupled system. The mathematical conventions, iteration design, experiment definitions, acceptance criteria, and scientific interpretation remain author-controlled; confidence in the computations is based on exact benchmarks, convergence studies, immutable characteristic data, and independent residual evaluation rather than on code generation itself.

Eight numerical experiments test complementary aspects of the method. The exact-solution tests cover Schwarzschild, Kerr, and Fisher–JNW spacetimes. They measure the metric error in regular regions, across a Schwarzschild horizon, and toward a curvature singularity. The vacuum experiments evolve a strong nonspherical outgoing perturbation and crossed characteristic data whose initial curvature is singular at the corner. The final experiments evolve nonspherical Einstein–scalar data and a stronger scalar pulse toward a trapped region. In the last experiment, an inner atlas of characteristic rectangles covers a curved domain adapted to null focusing. Independent four-metric evaluation finds the section

$$(u, v) = (-0.4696271025, 0.04) \tag{1.5}$$

on which the two future null expansions are strictly negative everywhere, with respective spherical suprema -0.0802152 and -4.28011 . Solving the angular marginally outer trapped surface equation on successive incoming null cones then reconstructs an apparent-horizon tube of 18 sections over $0.00441 \leq v \leq 0.05$. Coordinate and angular controls agree on its $v = 0.04$ section to better than 1.6×10^{-7} in the graph location.

This trapped-section computation is motivated by An’s proof that anisotropic outgoing characteristic perturbations of Christodoulou’s naked-singularity data generate an anisotropic apparent horizon that censors the singularity [1]. The numerical experiment does not constitute a proof of that theorem, and the reconstructed tube is tied to the chosen incoming-null foliation. It does, however, provide a coordinate- and angularly controlled numerical realization of the associated trapped-region mechanism and its anisotropic apparent horizon.

Section 2 fixes the double-null equations and records the analytic framework. Section 3 constructs the Picard map and its numerical discretization. Sections 4 and 5 present the vacuum and Einstein–scalar experiments, respectively, and distinguish exact metric errors from independently evaluated curvature residuals.

2 Equations and Theoretical Analysis

The null-frame decomposition of the Einstein equations and the associated geometric energy estimates were developed systematically in the proof of the nonlinear stability of Minkowski space [7]. Klainerman and Nicolò subsequently formulated the exterior evolution problem using a double-null foliation [15], while Christodoulou’s treatment of black-hole formation develops the optical structure equations, characteristic initial data, and null estimates in the same geometric setting [6]. This section fixes the particular conventions used in the numerical construction and then recalls the local existence framework relevant to the characteristic problem.

2.1 Double null foliation and equations

In this section, we introduce the equations for Lorentzian metric under double null foliation. Let $(\mathcal{M}, g)$ be a 4-dimensional Lorentzian manifold. With double null coordinates (u, v, θ^A) , we use (g_{AB}, b^A, Ω) to represent the metric (1.1). The level sets of u and v are denoted by H_u and $\underline{H}_v$ respectively, and their spherical intersection $H_u \cap \underline{H}_v$ is denoted by $S_{u,v}$. With double null frame

$$e_3 = \Omega^{-1}(\partial_u + b), \quad e_4 = \Omega^{-1}\partial_v, \quad e_A = \partial_{\theta^A},$$

we can define Ricci coefficients, or connection components

$$\begin{aligned} \chi_{AB} &= g(D_A e_4, e_B), \quad \underline{\chi}_{AB} = g(D_A e_3, e_B), \quad \eta_A = -\frac{1}{2}g(D_3 e_A, e_4), \\ \underline{\eta}_A &= -\frac{1}{2}g(D_4 e_A, e_3), \quad \omega = -\frac{1}{4}g(D_4 e_3, e_4), \quad \underline{\omega} = -\frac{1}{4}g(D_3 e_4, e_3), \\ \zeta_A &= \frac{1}{2}g(D_A e_4, e_3) = -\frac{1}{4}\Omega^{-2}(\mathcal{L}_v b^B)g_{AB}. \end{aligned} \quad (2.1)$$

Their transport equations are listed below:

$$\nabla_4 \operatorname{tr} \chi + |\chi|^2 = -\operatorname{Ric}_{44} - 2\omega \operatorname{tr} \chi, \quad \nabla_3 \operatorname{tr} \underline{\chi} + |\underline{\chi}|^2 = -\operatorname{Ric}_{33} - 2\underline{\omega} \operatorname{tr} \underline{\chi}, \quad (2.2)$$

$$\begin{aligned} \nabla_3 \hat{\chi}_{AB} + \frac{1}{2} \operatorname{tr} \chi \hat{\chi}_{AB} &= \widehat{\operatorname{Ric}}_{AB} + 2\underline{\omega} \hat{\chi}_{AB} + (\nabla \hat{\otimes} \eta)_{AB} + (\eta \hat{\otimes} \eta)_{AB} - \frac{1}{2} \operatorname{tr} \chi \hat{\chi}_{AB}, \\ \nabla_4 \hat{\underline{\chi}}_{AB} + \frac{1}{2} \operatorname{tr} \chi \hat{\underline{\chi}}_{AB} &= \widehat{\operatorname{Ric}}_{AB} + 2\omega \hat{\underline{\chi}}_{AB} + (\nabla \hat{\otimes} \underline{\eta})_{AB} + (\underline{\eta} \hat{\otimes} \underline{\eta})_{AB} - \frac{1}{2} \operatorname{tr} \chi \hat{\underline{\chi}}_{AB}, \end{aligned} \quad (2.3)$$

$$\begin{aligned} \nabla_3 \operatorname{tr} \chi &= -\operatorname{tr} \chi \operatorname{tr} \underline{\chi} + 2\underline{\omega} \operatorname{tr} \chi + 2\mathfrak{d}\mathfrak{v} \eta + 2|\eta|^2 - 2K + R + \operatorname{Ric}_{34}, \\ \nabla_4 \operatorname{tr} \underline{\chi} &= -\operatorname{tr} \chi \operatorname{tr} \underline{\chi} + 2\omega \operatorname{tr} \underline{\chi} + 2\mathfrak{d}\mathfrak{v} \underline{\eta} + 2|\underline{\eta}|^2 - 2K + R + \operatorname{Ric}_{34}. \end{aligned} \quad (2.4)$$

$$\begin{aligned} \nabla_4 \eta &= -\chi \cdot (\eta - \underline{\eta}) - \frac{1}{2} R_{A434}, \quad \nabla_3 \underline{\eta} = -\underline{\chi} \cdot (\underline{\eta} - \eta) - \frac{1}{2} R_{A343}, \\ \nabla_4 \underline{\omega} &= 2\omega \underline{\omega} - \frac{1}{2} K + \frac{1}{2} \operatorname{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \frac{1}{2} |\eta|^2 - \eta \cdot \underline{\eta}, \\ \nabla_3 \omega &= 2\omega \underline{\omega} - \frac{1}{2} K + \frac{1}{2} \operatorname{Ric}_{34} + \frac{1}{4} R + \frac{1}{4} \hat{\chi} \cdot \hat{\underline{\chi}} - \frac{1}{8} \operatorname{tr} \chi \operatorname{tr} \underline{\chi} + \frac{1}{2} |\underline{\eta}|^2 - \eta \cdot \underline{\eta}. \end{aligned} \quad (2.5)$$

Moreover, we have equations for ζ :

$$\begin{aligned}
\Omega \operatorname{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \operatorname{tr} \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta + \Omega \operatorname{tr} \underline{\chi} \nabla \log \Omega \\
&\quad + 2 \nabla(\Omega \underline{\omega}) + \operatorname{div}(\Omega \widehat{\underline{\chi}}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \underline{\chi}), \\
\Omega \operatorname{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \operatorname{tr} \chi \zeta - \Omega \widehat{\chi} \cdot \zeta + \Omega \operatorname{tr} \chi \nabla \log \Omega \\
&\quad + 2 \nabla(\Omega \omega) + \operatorname{div}(\Omega \widehat{\chi}) - \frac{1}{2} \nabla(\Omega \operatorname{tr} \chi).
\end{aligned} \tag{2.6}$$

For Weyl curvature,

$$W_{\mu\nu\theta\lambda} = R_{\mu\nu\theta\lambda} - \frac{1}{2} (\operatorname{Ric}_{\mu\theta} g_{\nu\lambda} + g_{\mu\theta} \operatorname{Ric}_{\nu\lambda} - \operatorname{Ric}_{\nu\theta} g_{\mu\lambda} - g_{\nu\theta} \operatorname{Ric}_{\mu\lambda}) + \frac{R}{6} (g_{\mu\theta} g_{\nu\lambda} - g_{\nu\theta} g_{\mu\lambda}),$$

we define the curvature components

$$\begin{aligned}
\beta_A &= \frac{1}{2} W(e_A, e_4, e_3, e_4), \quad \underline{\beta}_A = \frac{1}{2} W(e_A, e_3, e_3, e_4), \\
\rho &= \frac{1}{4} W(e_4, e_3, e_4, e_3), \quad \sigma = \frac{1}{4} {}^* W(e_4, e_3, e_4, e_3).
\end{aligned} \tag{2.7}$$

Then Gauss-Codazzi equations imply the following algebraic relations,

$$K = -\rho + \frac{1}{3} R + \frac{1}{2} \operatorname{Ric}_{34} + \frac{1}{2} \widehat{\underline{\chi}} \cdot \widehat{\chi} - \frac{1}{4} \operatorname{tr} \underline{\chi} \operatorname{tr} \chi, \tag{2.8}$$

$$\operatorname{curl} \eta = -\operatorname{curl} \underline{\eta} = \sigma + \frac{1}{2} \widehat{\underline{\chi}} \wedge \widehat{\chi}, \tag{2.9}$$

$$\operatorname{div} \widehat{\chi} - \frac{1}{2} \nabla \operatorname{tr} \chi = -\beta + \frac{1}{2} \operatorname{Ric}_4 + \frac{1}{2} \operatorname{tr} \chi \zeta - \zeta \cdot \widehat{\chi}, \tag{2.10}$$

$$\operatorname{div} \widehat{\underline{\chi}} - \frac{1}{2} \nabla \operatorname{tr} \underline{\chi} = \underline{\beta} + \frac{1}{2} \operatorname{Ric}_3 - \frac{1}{2} \operatorname{tr} \underline{\chi} \zeta + \zeta \cdot \widehat{\underline{\chi}}. \tag{2.11}$$

After introducing the reduced curvature

$$\beta_A^r = \beta_A - \frac{1}{2} \operatorname{Ric}_{4A}, \quad \underline{\beta}_A^r = \underline{\beta}_A + \frac{1}{2} \operatorname{Ric}_{3A}, \quad \sigma^r = \sigma + \frac{1}{2} \widehat{\underline{\chi}} \wedge \widehat{\chi}, \tag{2.12}$$

we can write the equations for curvature components

$$\begin{aligned}
\Omega \nabla_3 K + \Omega \operatorname{tr} \underline{\chi} K &= \operatorname{div}(\Omega \underline{\beta}^r) + \frac{1}{2} \operatorname{div}(\eta \Omega \operatorname{tr} \underline{\chi}) - \operatorname{div}(\eta \cdot \Omega \widehat{\underline{\chi}}), \\
\Omega \nabla_4 K + \Omega \operatorname{tr} \chi K &= -\operatorname{div}(\Omega \beta^r) + \frac{1}{2} \operatorname{div}(\underline{\eta} \Omega \operatorname{tr} \chi) - \operatorname{div}(\underline{\eta} \cdot \Omega \widehat{\chi}),
\end{aligned} \tag{2.13}$$

$$\begin{aligned}
\nabla_3 \beta^r + (\operatorname{tr} \underline{\chi} - 2 \underline{\omega}) \beta^r &= -\nabla K + {}^* \nabla \sigma + 2 \widehat{\underline{\chi}} \cdot \underline{\beta} \\
&\quad + \frac{1}{2} (\nabla(\widehat{\underline{\chi}} \cdot \widehat{\underline{\chi}}) - {}^* \nabla(\widehat{\underline{\chi}} \wedge \widehat{\underline{\chi}})) - \frac{1}{4} \nabla(\operatorname{tr} \chi \operatorname{tr} \underline{\chi}) \\
&\quad + 3(\eta \rho + {}^* \eta \sigma) + \nabla_A (g^{CD} \operatorname{Ric}_{CD}) - \nabla^B \operatorname{Ric}_{BA} - \frac{1}{2} \operatorname{tr} \underline{\chi} \operatorname{Ric}_{4A}, \\
\nabla_4 \underline{\beta}^r + (\operatorname{tr} \chi - 2 \omega) \underline{\beta}^r &= \nabla K + {}^* \nabla \sigma + 2 \widehat{\underline{\chi}} \cdot \underline{\beta} \\
&\quad - \frac{1}{2} (\nabla(\widehat{\underline{\chi}} \cdot \widehat{\underline{\chi}}) + {}^* \nabla(\widehat{\underline{\chi}} \wedge \widehat{\underline{\chi}})) + \frac{1}{4} \nabla(\operatorname{tr} \chi \operatorname{tr} \underline{\chi}) \\
&\quad - 3(\underline{\eta} \rho - {}^* \underline{\eta} \sigma) - \nabla_A (g^{CD} \operatorname{Ric}_{CD}) + \nabla^B \operatorname{Ric}_{BA} + \frac{1}{2} \operatorname{tr} \chi \operatorname{Ric}_{3A}.
\end{aligned} \tag{2.14}$$

Using (2.5), we derive

$$\begin{aligned}\Omega \nabla_3 \operatorname{curl} \underline{\eta} + \Omega \operatorname{tr} \underline{\chi} \operatorname{curl} \underline{\eta} &= \operatorname{curl}(\Omega \underline{\beta}^r) + \epsilon^{AC} \nabla_A (\Omega \underline{\chi} \cdot \underline{\eta})_C - \epsilon^{AC} \nabla_A (\Omega \operatorname{Ric})_{C3}, \\ \Omega \nabla_4 \operatorname{curl} \eta + \Omega \operatorname{tr} \chi \operatorname{curl} \eta &= -\operatorname{curl}(\Omega \beta^r) + \epsilon^{AC} \nabla_A (\Omega \chi \cdot \underline{\eta})_C - \epsilon^{AC} \nabla_A (\Omega \operatorname{Ric})_{C4},\end{aligned}\quad (2.15)$$

and

$$\begin{aligned}\Omega \nabla_3 (\operatorname{div} \underline{\eta} - K) + \Omega \operatorname{tr} \underline{\chi} (\operatorname{div} \underline{\eta} - K) &= 4 \operatorname{div}(\Omega \widehat{\underline{\chi}} \cdot \zeta) - \operatorname{div}(\Omega \operatorname{Ric}_3), \\ \Omega \nabla_4 (\operatorname{div} \eta - K) + \Omega \operatorname{tr} \chi (\operatorname{div} \eta - K) &= -4 \operatorname{div}(\Omega \widehat{\chi} \cdot \zeta) - \operatorname{div}(\Omega \operatorname{Ric}_4).\end{aligned}\quad (2.16)$$

For a scalar function ϕ , there is identity

$$-\Omega^2 \square \phi = \Omega e_3(\Omega e_4 \phi) + \frac{1}{2} \Omega \operatorname{tr} \chi \Omega e_3 \phi + \frac{1}{2} \Omega \operatorname{tr} \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi. \quad (2.17)$$

2.2 Local existence

We now introduce some results on Einstein vacuum equations. Similar results can be derived for other Einstein field equations and we omit them due to length of the article.

For data prescribed on two transversely intersecting null hypersurfaces, Rendall reduced the characteristic problem to a standard Cauchy problem and proved existence in a neighborhood of the intersection [19]. Luk then extended the solution to a neighborhood of the full initial null hypersurfaces by estimates adapted to a double-null foliation [16]. Related existence results have since been obtained for nonlinear symmetric hyperbolic systems, including Einstein equations with sources [5], and in the Newman–Penrose formalism in Stewart’s gauge [14]. We state the vacuum theorem in the form used here, following [16].

Theorem 2.1. *Given initial null hypersurface $H_0, \underline{H}_0$ and regular initial metric (g, Ω, b) on it, we define $\chi_{AB} = \frac{1}{2} \mathcal{L}_{e_4} g_{AB}$ on H_0 and $\underline{\chi}_{AB} = \frac{1}{2} \mathcal{L}_{e_3} g_{AB}$ on $\underline{H}_0$. Let ζ be a vector field on $H_0 \cap \underline{H}_0$. Suppose*

$$e_4(\operatorname{tr} \chi) + |\chi|^2 + 2\omega \operatorname{tr} \chi = 0, \quad e_3(\operatorname{tr} \underline{\chi}) + |\underline{\chi}|^2 + 2\underline{\omega} \operatorname{tr} \underline{\chi} = 0,$$

hold on $H_0, \underline{H}_0$ respectively, then there is a future open neighborhood of $H_0 \cup \underline{H}_0$ and metric (g, b, Ω) on it solving the Einstein vacuum equations and extending the initial data (g, Ω, b) and ζ .

Thus local existence extends along every portion of the initial null hypersurfaces on which the required norms remain controlled. Extension to a larger rectangle, such as $(0, 1)^2 \times \mathbb{S}^2$, requires corresponding a priori bounds throughout that region.

2.3 Residual control

In this paper, we target to deal with asymmetric Einstein equations, which in general do not have explicit solutions. To measure the computational error, we use Ricci residues. For Einstein equation

$$G_{\mu\nu} := \operatorname{Ric}_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = T_{\mu\nu},$$

we measure $\|G - T\|_{L^2(S_{u,v})}$ to evaluate the approximation. The theorems below reveals that in our case the difference of metrics is controlled by the difference of curvature residues.

Theorem 2.2. *We consider metric $\overset{\circ}{g}_{AB}, \overset{\circ}{b}, \overset{\circ}{\Omega}$ on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ such that*

$$\overset{\circ}{\operatorname{Ric}}_{44}(0, v) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{33}(u, 0) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{4A}(0, v) = 0, \quad \overset{\circ}{\operatorname{Ric}}_{3A}(u, 0) = 0,$$

and estimates

$$\|\mathring{\text{Ric}}_{AB}, \mathring{\text{Ric}}_3, \mathring{\text{Ric}}_4, \mathring{\text{Ric}}_{34}, \mathring{\text{Ric}}_{33}, \mathring{\text{Ric}}_{44}\|_{L_u^\infty L_v^\infty H^3(S)} \leq \epsilon. \quad (2.18)$$

We assume moreover

$$\|\partial_{\mathbb{S}} \mathring{g}, \partial_{\mathbb{S}} \mathring{b}, \partial_{\mathbb{S}} \mathring{\Omega}, \psi\|_{L_u^\infty L_v^\infty H^4(S)} \leq C_0, \quad (2.19)$$

where ψ stands for Ricci coefficients in interest. Then for $\epsilon = \epsilon(C_0)$ sufficiently small, there exists (g, b, Ω) solving EVE on $(0, 1) \times (0, 1) \times \mathbb{S}^2$ with initial data

$$(g, b, \Omega) = (\mathring{g}, \mathring{b}, \mathring{\Omega}), \text{ on } \underline{H}_0 \cup H_0,$$

and estimates

$$\|g - \mathring{g}, b - \mathring{b}, \Omega - \mathring{\Omega}\|_{L_u^\infty L_v^\infty H^4(S)} \leq \epsilon^{1/2}. \quad (2.20)$$

The methods of deriving the equations of difference and doing estimates are similar to section 7 of [3], so we only write the proof in sketch.

Define auxiliary functions $\omega^\dagger, \underline{\omega}^\dagger$ by

$$\nabla_3 \omega^\dagger = \frac{1}{2} \text{curl } \zeta, \quad \omega^\dagger|_{H_0} = 0, \quad \nabla_4 \underline{\omega}^\dagger = \frac{1}{2} \text{curl } \zeta, \quad \underline{\omega}^\dagger|_{\underline{H}_0} = 0.$$

We make bootstrap assumption

$$\begin{aligned} & \|\widetilde{\beta}, \widetilde{K}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_v^2 H^3(S)} \\ & + \|\widetilde{K}, \widetilde{\beta}, \nabla \widetilde{\eta}, \nabla \widetilde{\chi}, \nabla \widetilde{\omega}, \nabla \widetilde{\omega}^\dagger\|_{L_u^2 H^3(S)} \leq B \exp(D(v+u))\epsilon. \end{aligned}$$

We first fix B sufficiently large to control the initial-data and forcing terms, and then choose D sufficiently large to improve the bootstrap bound from B to $B/2$.

For motivation, let $f \geq 0$ satisfy $\partial_u f \leq C_0 f$ and $\partial_v f \leq C_0 f$. Assume its data on $u = 0$ and $v = 0$ are bounded by $C_0 \epsilon$. Under the bootstrap bound

$$f(u, v) \leq B \exp(D(v+u))\epsilon,$$

integration gives

$$\begin{aligned} f(u, v) - f(0, v) & \leq C_0 \int_0^u f(u', v) du' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Dv)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon, \\ f(u, v) - f(u, 0) & \leq C_0 \int_0^v f(u, v') dv' \\ & \leq \frac{C_0 B}{D} (\exp(D(u+v)) - \exp(Du)) \epsilon \leq \frac{C_0 B}{D} \exp(D(u+v)) \epsilon. \end{aligned}$$

Choosing $B \geq 4C_0$ and $D \geq 4C_0$ improves this bound to $(B/2) \exp(D(u+v))\epsilon$. With B and D fixed independently of ϵ , the resulting bound is $O(\epsilon) = o(\epsilon^{1/2})$ on a fixed finite region.

For the energy estimates we use the following integration bounds. All null integrals run from 0 to the indicated endpoint, and the hypotheses hold throughout the rectangle.

Lemma 2.3. *Let $D > 0$. Suppose $\|f\|_{L^2(S_{u,v})} \leq B \exp(D(v+u))\epsilon$, then we have*

$$\|f\|_{L_u^2 L^2(S)} + \|f\|_{L_v^2 L^2(S)} \leq \sqrt{2} B D^{-1/2} \exp(D(v+u))\epsilon. \quad (2.21)$$

Suppose either $\|h\|_{L_u^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$ or $\|h\|_{L_v^2 L^2(S)} \leq B \exp(D(u+v))\epsilon$, then we have

$$\|h\|_{L_u^2 L_v^2 L^2(S)} \leq B(2D)^{-1/2} \exp(D(v+u))\epsilon. \quad (2.22)$$

The same estimates hold with $H^3(S)$ in place of $L^2(S)$.

We use notation

$$\psi_1 \in \{\eta, \underline{\eta}, \chi, \omega\}, \quad \psi_2 \in \{\eta, \underline{\eta}, \underline{\chi}, \underline{\omega}\}$$

Because

$$\mathcal{L}_v b = -2\Omega^2(\eta - \underline{\eta}), \quad \mathcal{L}_v g = 2\Omega\chi, \quad \mathcal{L}_v \log \Omega = -2\Omega\omega,$$

we obtain that

$$\|\widetilde{b}, \widetilde{g}, \widetilde{\log \Omega}\|_{H^4(S_{u,v})} \lesssim \|\widetilde{\eta}, \widetilde{\underline{\eta}}, \widetilde{\chi}, \widetilde{\omega}, \widetilde{\Omega}, \widetilde{g}\|_{L_v^2 H^4(S)} \lesssim B \exp(D(u+v))\epsilon. \quad (2.23)$$

For Bianchi pair (Ψ_1, Ψ_2) in interest, we consider equations

$$\begin{aligned} \Omega \nabla_3 \widetilde{\Psi}_1 + \mathcal{D} \widetilde{\Psi}_2 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{b} \cdot \nabla \overset{\circ}{\Psi}_1 + \widetilde{\partial} g \overset{\circ}{\Psi}_2 + \widetilde{g} \nabla \overset{\circ}{\Psi}_2 + \overset{\circ}{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}, \\ \Omega \nabla_4 \widetilde{\Psi}_2 - {}^* \mathcal{D} \widetilde{\Psi}_1 &= \widetilde{\psi} \widetilde{\Psi} + \widetilde{\psi} \nabla \widetilde{\psi} + \widetilde{\psi}^3 + \widetilde{\partial} g \overset{\circ}{\Psi}_1 + \widetilde{g} \nabla \overset{\circ}{\Psi}_1 + \overset{\circ}{\psi} \overset{\circ}{\text{Ric}} + \nabla \overset{\circ}{\text{Ric}}. \end{aligned}$$

Energy estimates give that

$$\begin{aligned} \|\widetilde{\Psi}_1\|_{L_v^2 H^3(S)}^2 + \|\widetilde{\Psi}_2\|_{L_u^2 H^3(S)}^2 \\ \lesssim \epsilon^2 + \|\widetilde{\partial} g, \widetilde{g}, \widetilde{\psi}, \widetilde{\nabla \psi}, \widetilde{\Psi}, \widetilde{b}\|_{L_u^2 L_v^2 H^3(S)}^2 \lesssim (1 + B^2/D) \exp(2D(u+v))\epsilon^2. \end{aligned}$$

Applying the estimate to pairs

$$(\beta, (K, \text{curl } \eta)), \quad ((K, \text{curl } \underline{\eta}), \underline{\beta}), \quad (\nabla \text{tr } \underline{\chi}, 0), \quad (0, \nabla \text{tr } \chi),$$

$$(\text{div } \underline{\eta} - K, 0), \quad (0, \text{div } \eta - K), \quad (\nabla \omega + {}^* \nabla \omega^\dagger - \frac{1}{2} \beta, 0), \quad (0, -\nabla \underline{\omega} + {}^* \nabla \underline{\omega}^\dagger - \frac{1}{2} \underline{\beta}),$$

yields a bound $C(1 + B^2/D) \exp(2D(u+v))\epsilon^2$ for the square of the full bootstrap norm, where $C = C(C_0)$ is independent of B , D , and ϵ . Choosing $B^2 \geq 8C$ and then $D \geq 8C$ improves the bootstrap bound to $(B/2) \exp(D(u+v))\epsilon$, completing the proof sketch.

Remark 2.4. The conditions

$$\overset{\circ}{\text{Ric}}_{44}(0, v) = 0, \quad \overset{\circ}{\text{Ric}}_{33}(u, 0) = 0,$$

guarantee that $(\overset{\circ}{g}, \overset{\circ}{b}, \overset{\circ}{\Omega})$ is a valid initial data set for characteristic initial value problem of Einstein vacuum equation, and

$$\overset{\circ}{\text{Ric}}_{4A}(0, v) = 0, \quad \overset{\circ}{\text{Ric}}_{3A}(u, 0) = 0,$$

ensure that $\overset{\circ}{\zeta}$ along H_0 and $\underline{H}_0$ is equal to the ζ of corresponding genuine solution.

3 Picard Iteration Design and Numerical Strategies

In this section, we introduce the iteration design and numerical strategies. The Picard hierarchy below is a numerical adaptation of the approximation scheme in [2]. We preserve its principal dependency order—freezing nonlinear coefficients at the previous iterate and successively solving transport equations in their geometrically distinguished null directions—but modify individual equations when an equivalent form is more suitable for numerical integration. Accordingly, the formulas in this section define the numerical iteration used in the present work; they should not be read as a literal transcription of the analytic scheme in [2].

3.1 Picard iteration for Einstein equations

In this section, we first consider Einstein scalar-field equations

$$\text{Ric} = d\phi \otimes d\phi, \quad \square_g \phi = 0.$$

We work in region

$$Q = \{-1 < u < u_0, \quad 0 < v < v_0(u)\}.$$

The first task is to construct a valid initial data for ESE. We directly prescribe g, b, Ω along $\underline{H}_0$. The only requirement is

$$-\Omega \nabla_3(\Omega \text{tr } \underline{\chi}) - \frac{1}{2}(\Omega \text{tr } \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \text{tr } \underline{\chi} \geq 0$$

Then $\Omega e_3 \phi(u, 0)$ can be assigned via equation

$$\Omega e_3 \phi = \left(-\Omega \nabla_3(\Omega \text{tr } \underline{\chi}) - \frac{1}{2}(\Omega \text{tr } \underline{\chi})^2 - |\Omega \widehat{\underline{\chi}}|^2 - 4\Omega \underline{\omega} \Omega \text{tr } \underline{\chi} \right)^{1/2}.$$

After defining $\phi(-1, 0)$ and $\zeta(-1, 0)$, we derive $\phi(u, 0)$ from $\Omega e_3 \phi(u, 0)$ and $\zeta(u, 0)$ via

$$\Omega \nabla_3 \zeta + \frac{3}{2} \Omega \text{tr } \underline{\chi} \zeta + \Omega \widehat{\underline{\chi}} \cdot \zeta = -2\nabla(\Omega \underline{\omega}) - \text{div}(\Omega \widehat{\underline{\chi}}) + \frac{1}{2} \nabla(\Omega \text{tr } \underline{\chi}) - \Omega \text{tr } \underline{\chi} \nabla \log \Omega + \Omega e_3 \phi \nabla \phi.$$

Thus $\eta = \zeta + \nabla \log \Omega$ and $\eta = -\zeta + \nabla \log \Omega$ are also prescribed. We then define $\Omega \text{tr } \chi(-1, 0)$ and $\Omega \widehat{\chi}(-1, 0)$, and compute $\Omega \chi(u, 0)$ by

$$\Omega \nabla_3(\Omega \text{tr } \chi) + \Omega \text{tr } \underline{\chi} \Omega \text{tr } \chi = \Omega^2 \left(2 \text{div } \eta + 2 |\eta|^2 - 2K(g) + |\nabla \phi|^2 \right),$$

$$\Omega \nabla_3(\Omega \widehat{\chi}) + \frac{1}{2} \Omega \text{tr } \underline{\chi} \Omega \widehat{\chi} = \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) - \frac{1}{2} \Omega \text{tr } \chi \Omega \widehat{\underline{\chi}} + \frac{1}{2} \Omega^2 \nabla \phi \hat{\otimes} \nabla \phi.$$

We also directly assign $\Omega e_4 \phi(-1, 0)$ and $\Omega \omega(-1, 0)$, and compute them along $\underline{H}_0$ via

$$\Omega e_3(\Omega e_4 \phi) + \frac{1}{2} \Omega \text{tr } \chi \Omega e_3 \phi + \frac{1}{2} \Omega \text{tr } \underline{\chi} \Omega e_4 \phi - \Omega^2 \Delta \phi - 2\Omega^2 \eta \cdot \nabla \phi = 0,$$

$$\frac{1}{2} \Omega e_3 \phi \Omega e_4 \phi + \frac{1}{2} \Omega^2 |\nabla \phi|^2 = 2\Omega \nabla_3(\Omega \omega) + \Omega^2 K - \frac{1}{2} \Omega \widehat{\underline{\chi}} \cdot \Omega \widehat{\underline{\chi}} + \frac{1}{4} \Omega \text{tr } \chi \Omega \text{tr } \underline{\chi} - \Omega^2 |\underline{\eta}|^2 + 2\Omega^2 \eta \cdot \underline{\eta}.$$

Along H_{-1} , we prescribe $\Omega e_4 \phi(-1, v)$ and $\Omega \omega(-1, v)$ to compute ϕ, Ω . Consider a symmetric and $g(-1, 0)$ -trace-free tensor $X_{AB}(-1, v)$. We then prescribe $g, \Omega \chi$ along H_{-1} by solving:

$$\begin{cases} \Omega \widehat{\chi}_{AB} = \frac{1}{2} \left(X_{AC} (g(-1, 0))^{CD} g_{DB} + X_{BC} (g(-1, 0))^{CD} g_{DA} \right), \\ \partial_v(\Omega \text{tr } \chi) = -\frac{1}{2}(\Omega \text{tr } \chi)^2 - 4\Omega \omega \Omega \text{tr } \chi - (\Omega e_4 \phi)^2 - \Omega \widehat{\chi}_{AB} \Omega \widehat{\chi}_{CD} g^{AC} g^{BD}, \\ \mathcal{L}_v g_{AB} = \Omega \text{tr } \chi g_{AB} + 2\Omega \widehat{\chi}_{AB}. \end{cases} \quad (3.1)$$

With these initial data, we can start the iteration. The quantities to be iterated are

$$g^{(i)}, (b^{(i)})^A, \log \Omega^{(i)}, (\Omega \chi)_{AB}^{(i)}, (\Omega \underline{\chi})_{AB}^{(i)}, (\zeta^{(i)})^A, (\Omega \omega)^{(i)}, (\Omega \underline{\omega})^{(i)}, (\Omega e_3 \phi)^{(i)}, (\Omega e_4 \phi)^{(i)}, \nabla_A \phi^{(i)}.$$

Initially, we set $g^{(0)} = g|_{v=0}$, $\Omega^{(0)} = \Omega|_{v=0}$, $\zeta^{(0)} = \zeta|_{v=0}$,

$$(b^{(0)})^A(u, v) = b^A(u, 0) - \int_0^v 4(\Omega^{(0)})^2 (\zeta^{(0)})^A(u, v') \, dv',$$

$\widehat{\chi}^{(0)} = \widehat{\chi}|_{v=0}$, $\text{tr } \chi^{(0)} = \text{tr } \chi|_{v=0}$, $\widehat{\underline{\chi}}^{(0)} = \widehat{\underline{\chi}}|_{v=0}$, $\text{tr } \underline{\chi}^{(0)} = \text{tr } \underline{\chi}|_{v=0}$, $\omega^{(0)} = \omega|_{v=0}$, $\underline{\omega}^{(0)} = \underline{\omega}|_{v=0}$, $(e_3\phi)^{(0)} = e_3\phi|_{v=0}$, $(e_4\phi)^{(0)} = e_4\phi|_{v=0}$, $(\nabla_A\phi)^{(0)} = \nabla_A\phi|_{v=0}$. For all i , we always require

$$\eta_A^{(i)} = (\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}, \quad \underline{\eta}_A^{(i)} = -(\zeta^{(i)})^B g_{AB}^{(i)} + \nabla_A \log \Omega^{(i)}$$

1. Construction of $(\Omega\widehat{\chi})^{(i+1/2)}$, $\partial_v\phi^{(i+1)}$, and $\nabla_A\phi^{(i+1)}$

We use (2.3) to define $(\Omega\widehat{\chi})^{(i+1/2)}$,

$$\left\{ \begin{aligned} & \mathcal{L}_{\partial_u+b^{(i)}}(\Omega\widehat{\chi}^{(i+1/2)})_{AB} - \frac{1}{2}(\Omega \text{tr } \underline{\chi})^{(i)}(\Omega\widehat{\chi}^{(i+1/2)})_{AB} \\ & \quad - (\Omega\widehat{\underline{\chi}}^{(i)})_A{}^C(\Omega\widehat{\chi}^{(i+1/2)})_{CB} - (\Omega\widehat{\underline{\chi}}^{(i)})_B{}^C(\Omega\widehat{\chi}^{(i+1/2)})_{CA} \\ & = (\Omega^{(i)})^2 \left(\left(\nabla^{(i)} \hat{\otimes} \eta^{(i)} + \eta^{(i)} \hat{\otimes} \nabla^{(i)} + \frac{1}{2} \nabla\phi^{(i)} \hat{\otimes} \nabla\phi^{(i)} \right)_{AB} - \frac{1}{2}(\text{tr } \chi)^{(i)} \widehat{\underline{\chi}}_{AB}^{(i)} \right), \\ & \Omega\widehat{\chi}^{(i+1/2)}_{AB}(-1, v) = \frac{1}{2} \left(\Omega\widehat{\chi}_{AC} g^{CD} g^{(i)}_{DB} + \Omega\widehat{\chi}_{BC} g^{CD} g^{(i)}_{DA} \right) (-1, v). \end{aligned} \right. \quad (3.2)$$

Using (2.17), we define $\partial_v\phi^{(i+1)}$ by

$$\left\{ \begin{aligned} & (\Omega e_3)^{(i)}(\partial_v\phi^{(i+1)}) + \frac{1}{2}(\Omega \text{tr } \underline{\chi})^{(i)}(\partial_v\phi^{(i+1)}) \\ & = (\Omega^{(i)})^2 \Delta^{(i)}\phi^{(i)} - \frac{1}{2}(\Omega \text{tr } \chi)^{(i)}(\Omega e_3)^{(i)}\phi^{(i)} + 2(\Omega^{(i)})^2 \eta^{(i)} \nabla\phi^{(i)}, \\ & \partial_v\phi^{(i+1)}(-1, v) = \partial_v\phi(-1, v). \end{aligned} \right. \quad (3.3)$$

It follows that

$$\nabla_A\phi^{(i+1)}(u, v) = \nabla_A\phi(u, 0) + \partial_A \left(\int_0^v \partial_v\phi^{(i+1)} \right).$$

2. Construction of $(\Omega\underline{\omega})^{(i+1/2)}$

We use (2.5) to define $(\Omega\underline{\omega})^{(i+1/2)}$,

$$\left\{ \begin{aligned} & 4\partial_v(\Omega\underline{\omega})^{(i+1/2)} = \Omega\widehat{\chi}_{AB}^{(i+1/2)}\Omega\widehat{\underline{\chi}}_{CD}^{(i)}(g^{(i)})^{AC}(g^{(i)})^{BD} + \frac{1}{2}(\Omega \text{tr } \chi)^{(i)}(\Omega \text{tr } \underline{\chi})^{(i)} \\ & \quad - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \text{tr } \chi)^{(i)} - 2(\Omega^{(i)})^2 \text{div}^{(i)} \eta^{(i)} \\ & \quad + (\partial_v\phi)^{(i+1)}(\Omega e_3\phi)^{(i)}, \\ & (\Omega\underline{\omega})^{(i+1/2)}(u, 0) = \Omega\underline{\omega}(u, 0). \end{aligned} \right. \quad (3.4)$$

3. Construction of $\Omega^{(i+1)}$, $\nabla_A \log \Omega^{(i+1)}$, and $(\Omega\omega)^{(i+1)}$

With $(\Omega\underline{\omega})^{(i+1/2)}$, we can integrate it to obtain $\Omega^{(i+1)}$

$$\left\{ \begin{aligned} & \left(\partial_u + b^{(i)} \right) \log \Omega^{(i+1)} = -2(\Omega\underline{\omega})^{(i+1/2)}, \\ & \log \Omega^{(i+1)}(-1, v) = \log \Omega(-1, v), \end{aligned} \right. \quad (3.5)$$

$$\nabla_A \log \Omega^{(i+1)} = \partial_A \log \Omega^{(i+1)}. \quad (3.6)$$

We construct $(\Omega\omega)^{(i+1)}$ by integration instead of differentiation,

$$\left\{ \begin{aligned} & 4 \left(\partial_u + b^{(i)} \right) (\Omega\omega)^{(i+1)} = \Omega\widehat{\chi}_{AB}^{(i+1/2)}\Omega\widehat{\underline{\chi}}_{CD}^{(i)}(g^{(i)})^{AC}(g^{(i)})^{BD} + \frac{1}{2}(\Omega \text{tr } \chi)^{(i)}(\Omega \text{tr } \underline{\chi})^{(i)} \\ & \quad - 4(\Omega^{(i)})^2 \eta^{(i)} \cdot \underline{\eta}^{(i)} + (\partial_u + b^{(i)})(\Omega \text{tr } \chi)^{(i)} - 2(\Omega^{(i)})^2 \text{div}^{(i)} \eta^{(i)} \\ & \quad + (\partial_v\phi)^{(i+1)}(\Omega e_3\phi)^{(i)} - 8(\Omega^{(i)})^2 (\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}, \\ & (\Omega\omega)^{(i+1)}(-1, v) = \Omega\omega(-1, v). \end{aligned} \right. \quad (3.7)$$

We note that the construction of $(\Omega\omega)^{(i+1)}$ is equivalent to $(\Omega\omega)^{(i+1)} = -\frac{1}{2}\partial_v \log \Omega^{(i+1)}$, because

$$\begin{aligned} 4(\Omega e_3)^{(i)}(\Omega\omega)^{(i+1)} &= -2\partial_v(\Omega e_3)^{(i)} \log \Omega^{(i+1)} + 2\mathcal{L}_v b^{(i)} \nabla \log \Omega^{(i+1)} \\ &= 4\partial_v(\Omega\omega)^{(i+1/2)} - 8(\Omega^{(i)})^2(\zeta^{(i)})^A \nabla_A \log \Omega^{(i+1)}. \end{aligned} \quad (3.8)$$

4. Construction of $\zeta^{(i+1)}$, $b^{(i+1)}$, and $(\Omega\omega)^{(i+1)}$

From (2.6), we define $\zeta^{(i+1)}$,

$$\left\{ \begin{aligned} \mathcal{L}_v(\zeta^{(i+1)})^A &= -2(\Omega \operatorname{tr} \chi)^{(i)}(\zeta^{(i)})^A - 2(g^{(i)})^{AC} \left((\Omega\widehat{\chi})^{(i+1/2)} \right)_{CB} (\zeta^{(i)})^B \\ &\quad + 2(g^{(i)})^{AB} \partial_B (\Omega\omega)^{(i+1)} + (\operatorname{div})^{(i)}(\Omega\widehat{\chi})_B^{(i+1/2)} (g^{(i)})^{BA} \\ &\quad - \frac{1}{2}(g^{(i)})^{AB} \partial_B (\Omega \operatorname{tr} \chi)^{(i)} + (\Omega \operatorname{tr} \chi)^{(i)} (g^{(i)})^{AB} \partial_B \log \Omega^{(i+1)} \\ &\quad - (\partial_v \phi)^{(i+1)} \partial_B \phi^{(i)} (g^{(i)})^{AB}, \\ (\zeta^{(i+1)})^A(u, 0) &= \zeta^A(u, 0). \end{aligned} \right. \quad (3.9)$$

We remark that it is necessary to treat $\zeta^{(i+1)}$ as a vector field instead of a 1-form in order to define $b^{(i+1)}$,

$$\left\{ \begin{aligned} \mathcal{L}_v(b^{(i+1)})^A &= -4(\Omega^{(i+1)})^2(\zeta^{(i+1)})^A, \\ (b^{(i+1)})^A(u, 0) &= b^A(u, 0). \end{aligned} \right. \quad (3.10)$$

The incoming lapse coefficient must then use the updated shift:

$$(\Omega\omega)^{(i+1)} = (\Omega\omega)^{(i+1/2)} - \frac{1}{2}(b^{(i+1)} - b^{(i)}) \cdot \nabla \log \Omega^{(i+1)}. \quad (3.11)$$

The numerical implementation applies the angular projection to this correction and restores the prescribed traces after projection and relaxation.

5. Construction of $g^{(i+1)}$ and $(\Omega\chi)^{(i+1)}$

It remains to define $g^{(i+1)}$. We consider the following $(g, \operatorname{tr} \chi, \widehat{\chi})$ system,

$$\left\{ \begin{aligned} \mathcal{L}_v(g^{(i+1)})_{AB} &= (\Omega \operatorname{tr} \chi)^{(i+1)}(g^{(i+1)})_{AB} + 2(\Omega\widehat{\chi})_{AB}^{(i+1)}, \\ \partial_v (\Omega \operatorname{tr} \chi)^{(i+1)} &= -\frac{1}{2} \left((\Omega \operatorname{tr} \chi)^{(i+1)} \right)^2 - 4(\Omega\omega)^{(i+1)}(\Omega \operatorname{tr} \chi)^{(i+1)} \\ &\quad - (\Omega\widehat{\chi})_{AB}^{(i+1)} (\Omega\widehat{\chi})_{CD}^{(i+1)} g^{(i+1)AC} g^{(i+1)BD} - (\partial_v \phi^{(i+1)})^2, \\ (\Omega\widehat{\chi})_{AB}^{(i+1)} &= \frac{1}{2} \left((\Omega\widehat{\chi})_{AC}^{(i+1/2)} (g^{(i)})^{CD} g^{(i+1)}_{DB} + (\Omega\widehat{\chi})_{BC}^{(i+1/2)} (g^{(i)})^{CD} g^{(i+1)}_{DA} \right), \\ (g^{(i+1)})_{AB}(u, 0) &= g_{AB}(u, 0), \quad (\Omega \operatorname{tr} \chi)^{(i+1)}(u, 0) = \Omega \operatorname{tr} \chi(u, 0). \end{aligned} \right. \quad (3.12)$$

6. Construction of $(\Omega\chi)^{(i+1)}$ and $(\Omega e_3\phi)^{(i+1)}$

There are two equivalent ways to define $(\Omega\chi)^{(i+1)}$. The first one is using $\Omega\chi_{AB} = \frac{1}{2}\mathcal{L}_{\Omega e_3} g_{AB}$. We use the second way, which is using equation

$$\left\{ \begin{aligned} \mathcal{L}_v(\Omega\chi)_{AB}^{(i+1)} &= \mathcal{L}_{\partial_u + b^{(i+1)}}(\Omega\chi)_{AB}^{(i+1)} - 2(\Omega^{(i+1)})^2 \left(\nabla_A^{(i+1)} \zeta_B^{(i+1)} + \nabla_B^{(i+1)} \zeta_A^{(i+1)} \right) \\ &\quad - 4(\Omega^{(i+1)})^2 \left(\zeta_A^{(i+1)} \nabla_B \log \Omega^{(i+1)} + \zeta_B^{(i+1)} \nabla_A \log \Omega^{(i+1)} \right), \\ (\Omega\chi)_{AB}^{(i+1)}(u, 0) &= \Omega\chi_{AB}(u, 0). \end{aligned} \right. \quad (3.13)$$

We remark that

$$\begin{aligned} &\Omega \nabla_4 (\Omega\chi)_{AB} - \Omega \nabla_3 (\Omega\chi)_{AB} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - (\Omega\chi)_A^C (\Omega\chi)_{CB} - (\Omega\chi)_B^C (\Omega\chi)_{CA} \\ &\quad - \mathcal{L}_{\partial_u + b}(\Omega\chi)_{AB} + (\Omega\chi)_A^C (\Omega\chi)_{CB} + (\Omega\chi)_B^C (\Omega\chi)_{AC} \\ &= \mathcal{L}_v(\Omega\chi)_{AB} - \mathcal{L}_{\partial_u + b}(\Omega\chi)_{AB}. \end{aligned} \quad (3.14)$$

Theoretically, $(\Omega \underline{\chi})_{AB}^{(i+1)} = \frac{1}{2} \mathcal{L}_{\partial_u + b^{(i+1)}} g_{AB}^{(i+1)}$. Similarly, we construct $(\Omega e_3 \phi)^{(i+1)}$ by

$$\begin{cases} \partial_v (\Omega e_3 \phi)^{(i+1)} = (\partial_u + b^{(i+1)}) \partial_v \phi^{(i+1)} - 4(\Omega^{(i+1)})^2 (\zeta^{(i+1)})^A \nabla_A \phi^{(i+1)}, \\ (\Omega e_3 \phi)^{(i+1)}(u, 0) = \Omega e_3 \phi(u, 0). \end{cases} \quad (3.15)$$

For Einstein vacuum equation

$$\text{Ric}_{\mu\nu} = 0,$$

removing all ϕ -terms of ESE gives the iterative construction directly, so we omit the repetition of construction.

3.2 Curvature residues

Computing curvature usually involves second order derivatives, which would heavily reduce the numerical precision. Especially, $\lim_{v \rightarrow 0} R_{4A4B}(-1, v)$ can be infinite for short pulse perturbation and Ric_{44} is hard to compute in this way, even though it is theoretically zero in our construction. See [1] for the example of short pulse with infinite R_{4A4B} . We use intrinsic equations of Lorentzian manifold to compute the Ricci curvature:

$$\begin{aligned} \Omega^2 \text{Ric}_{33} &= -\Omega \nabla_3 (\Omega \text{tr} \underline{\chi}) - 4\Omega \underline{\omega} \Omega \text{tr} \underline{\chi} - \frac{1}{2} (\Omega \text{tr} \underline{\chi})^2 - |\Omega \widehat{\chi}|^2, \\ \Omega \text{Ric}_{3A} &= \Omega \nabla_3 \zeta + \frac{3}{2} \Omega \text{tr} \underline{\chi} \zeta + \Omega \widehat{\chi} \cdot \zeta + 2\nabla(\Omega \underline{\omega}) + \text{div}(\Omega \widehat{\chi}) - \frac{1}{2} \nabla(\Omega \text{tr} \underline{\chi}) + \Omega \text{tr} \underline{\chi} \nabla \log \Omega, \\ \Omega \text{Ric}_{4A} &= -\Omega \nabla_4 \zeta - \frac{3}{2} \Omega \text{tr} \chi \zeta - \Omega \widehat{\chi} \cdot \zeta + 2\nabla(\Omega \omega) + \text{div}(\Omega \widehat{\chi}) - \frac{1}{2} \nabla(\Omega \text{tr} \chi) + \Omega \text{tr} \chi \nabla \log \Omega, \\ \Omega^2 \widehat{\text{Ric}}_{AB} &= \Omega \nabla_3 (\Omega \widehat{\chi}) + \frac{1}{2} \Omega \text{tr} \underline{\chi} \Omega \widehat{\chi} - \Omega^2 (\nabla \hat{\otimes} \eta + \eta \hat{\otimes} \eta) + \frac{1}{2} \Omega \text{tr} \chi \Omega \widehat{\chi}, \\ \Omega^2 g^{AB} \text{Ric}_{AB} &= \Omega \nabla_3 (\Omega \text{tr} \chi) + \Omega \text{tr} \chi \Omega \text{tr} \underline{\chi} - 2\Omega^2 \text{div} \eta - 2\Omega^2 |\eta|^2 + 2\Omega^2 K, \\ \Omega^2 \text{Ric}_{34} &= 4\Omega \nabla_4 (\Omega \underline{\omega}) - \Omega \widehat{\chi} \cdot \Omega \widehat{\chi} - \frac{1}{2} \Omega \text{tr} \chi \Omega \text{tr} \underline{\chi} + 4\Omega^2 \eta \cdot \underline{\eta} - \Omega \nabla_3 (\Omega \text{tr} \chi) + 2\Omega^2 \text{div} \eta. \end{aligned}$$

For a fresh check of the outgoing constraint we also use

$$\Omega^2 \text{Ric}_{44} = -\partial_v (\Omega \text{tr} \chi) - 4(\Omega \omega)(\Omega \text{tr} \chi) - \frac{1}{2} (\Omega \text{tr} \chi)^2 - |\Omega \widehat{\chi}|^2.$$

The coefficient 4 includes the product-rule contribution from weighting $\text{tr} \chi$ by Ω , since $\partial_v \log \Omega = -2\Omega \omega$. Thus no outgoing null derivative of $\widehat{\chi}$ is required. These formulas use first null derivatives of the stored connection variables; the intrinsic Gauss curvature K uses the angularly smooth section metric. The numerical metric-connection relations are checked separately. Endpoint values at singular coordinate Jacobians are excluded from the reported residuals.

3.3 Legendre–Gauss–Lobatto spectral elements

In the experiments, we will add non-spherical symmetric perturbation to the Einstein scalar-field or vacuum systems. To make the metric less regular, the perturbation is required to be of size $O(v^\delta)$. To deal with the non-smoothness, the characteristic coordinates are first replaced by coordinates in which the corner profiles are smoother. In particular, we use

$$\tau = -\log(-u), \quad s = \left(\frac{v}{v_{\max}} \right)^\delta, \quad u = -e^{-\tau}, \quad v = v_{\max} s^{1/\delta}.$$

Thus a profile proportional to v^δ is linear in s . This change of variables does not regularize the physical solution; it only represents its fractional behavior by a smooth function of the computational coordinate. Physical derivatives and integrals are recovered by

$$\partial_u = \frac{1}{-u} \partial_\tau, \quad \partial_v = \left(\frac{dv}{ds} \right)^{-1} \partial_s, \quad \int_0^v F(v') dv' = \int_0^s F(v(s')) \frac{dv}{ds'} ds'.$$

We divide each computational coordinate interval into elements. On an element of degree p , let $\{\xi_j\}_{j=0}^p$ be the Legendre–Gauss–Lobatto nodes, characterized by

$$(1 - \xi_j^2) P'_p(\xi_j) = 0,$$

including the two endpoints. A component F is represented by its Lagrange interpolant

$$I_p F(\xi) = \sum_{j=0}^p F(\xi_j) \ell_j(\xi).$$

The element differentiation and indefinite-integration matrices are

$$D_{ij} = \ell'_j(\xi_i), \quad Q_{ij} = \int_{-1}^{\xi_i} \ell_j(\xi) d\xi.$$

Consequently, interpolation of intermediate Runge–Kutta stages, differentiation, and cumulative integration are all obtained from the same element polynomial. Neighboring elements share their endpoint value, while the two derivative traces at an interior interface are combined into a single interface derivative. This composite construction permits local refinement in either characteristic direction without replacing the underlying transport hierarchy.

3.4 Tensor spherical-harmonic Galerkin discretization

The angular variables are represented on a quasi-uniform, pole-free point set on $\mathbb{S}^2$. Geometric vectors and tensors are stored as ambient Cartesian tensors and projected to the tangent bundle. This avoids the coordinate singularities of a polar chart. For a scalar field, we use the real spherical-harmonic expansion

$$f_L(\vartheta) = \sum_{\ell=0}^L \sum_{m=-\ell}^{\ell} f_{\ell m} Y_{\ell m}(\vartheta).$$

The coefficients are obtained by an overdetermined least-squares analysis of the nodal values, and angular derivatives are computed by differentiating the harmonic basis.

Vectors and symmetric two-tensors are not expanded componentwise as unrelated scalars. We use the electric and magnetic vector harmonics

$$Y_A^{E,\ell m} \sim \nabla_A Y_{\ell m}, \quad Y_A^{B,\ell m} \sim \epsilon_A^B \nabla_B Y_{\ell m},$$

and the trace, electric, and magnetic tensor harmonics

$$\gamma_{AB} Y_{\ell m}, \quad Y_{AB}^{E,\ell m} \sim (\nabla_A \nabla_B Y_{\ell m})^{\text{TF}}, \quad Y_{AB}^{B,\ell m} \sim \epsilon_{(A}^C \nabla_{B)} \nabla_C Y_{\ell m}.$$

These typed spaces preserve tangency, symmetry, and the tensorial transformation law. Trace-free quantities are projected with respect to the current section metric,

$$T_{AB}^{\text{TF}} = T_{AB} - \frac{1}{2} (\text{tr}_\gamma T) \gamma_{AB},$$

rather than with respect to a fixed background metric.

Let $\mathcal{V}_L$ denote the retained harmonic space and let $\mathcal{V}_W$, with $W > L$, be a work space used to form angular products. For a semidiscrete unknown $U_L \in \mathcal{V}_L$, every nonlinear right-hand side is defined by

$$\partial U_L = \Pi_L F(U_L),$$

where $F(U_L)$ is evaluated in the work space and Π_L is the appropriate scalar, vector, or tensor Galerkin projection. The retained/work separation reduces angular aliasing while keeping the evolved state in one declared finite-dimensional geometric space.

4 Results for Einstein Vacuum Equations

We test the vacuum iteration in five regimes of increasing geometric and numerical difficulty. The first three experiments admit explicit reference metrics and therefore measure the error directly by comparing the computed section metric with the exact one. The last two experiments evolve genuinely nonspherical characteristic data, for which no explicit spacetime is available; there we report a positive norm of the first-order Ricci residual on a protected interior region, together with metric–connection consistency checks. The residual is evaluated from the stored weighted connection variables and fresh derivatives, with the audit grid specified for each case. This avoids second null derivatives of the metric and the potentially singular derivative $\nabla_4 \hat{\chi}$ at the initial corner; intrinsic angular curvature uses the smooth sphere data. These diagnostics assess the numerical equations and do not furnish a certified metric-error bound.

4.1 Regular Schwarzschild and Kerr benchmarks

We begin with the Schwarzschild and subextremal Kerr families. In Schwarzschild coordinates, the metric of mass $M > 0$ is

$$\mathbf{g}_{\text{Sch}} = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2). \quad (4.1)$$

Let

$$r_* = r + 2M \log \left(\frac{r}{2M} - 1 \right) + C, \quad t = u + v, \quad r_* = v - u, \quad (4.2)$$

where the additive constant C fixes the reference sphere. Then (4.1) becomes

$$\mathbf{g}_{\text{Sch}} = -4\Omega^2 du dv + \gamma_{AB} d\theta^A d\theta^B, \quad \Omega^2 = 1 - \frac{2M}{r}, \quad \gamma = r^2 \gamma^\circ, \quad b = 0, \quad (4.3)$$

with $r = r(v - u)$. Thus the Schwarzschild benchmark is already in the double-null gauge used by the iteration.

For Kerr, write a for the angular momentum per unit mass and set

$$\rho^2 = r^2 + a^2 \cos^2 \vartheta, \quad \Delta = r^2 - 2Mr + a^2. \quad (4.4)$$

The Boyer–Lindquist expression is

$$\begin{aligned} \mathbf{g}_{\text{K}} = & - \left(1 - \frac{2Mr}{\rho^2}\right) dt^2 - \frac{4Mar \sin^2 \vartheta}{\rho^2} dt d\varphi + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\vartheta^2 \\ & + \frac{\sin^2 \vartheta}{\rho^2} ((r^2 + a^2)^2 - a^2 \Delta \sin^2 \vartheta) d\varphi^2. \end{aligned} \quad (4.5)$$

To put (4.5) into double-null form, we use the optical coordinates $(u, v, \vartheta_*, \varphi_*)$ defined by

$$t = u + v, \quad s = u - v, \quad r = r(s, \vartheta_*), \quad \vartheta = \vartheta(s, \vartheta_*), \quad \varphi = \varphi_* + h(s, \vartheta_*). \quad (4.6)$$

The functions r , ϑ , and h are obtained from the Kerr optical map, with $r = 4M$, $\vartheta = \vartheta_*$, and $h = 0$ on the reference two-sphere. If $\tilde{g}_{\alpha\beta}$ denotes the pullback of (4.5) under (4.6), the optical equations give

$$\tilde{g}_{vv} = 0, \quad \tilde{g}_{vA} = 0, \quad \tilde{g}_{uu} = \gamma_{AB} b^A b^B. \quad (4.7)$$

Consequently the pulled-back metric is exactly

$$\begin{aligned} \mathbf{g}_K &= -4\Omega^2 du dv + \gamma_{AB} (d\theta^A - b^A du)(d\theta^B - b^B du), \\ \Omega^2 &= -\frac{1}{2}\tilde{g}_{uv}, \quad \gamma_{AB} = \tilde{g}_{AB}, \quad b^A = -\gamma^{AB}\tilde{g}_{uB}. \end{aligned} \quad (4.8)$$

All runs use $M = 1$. We take $a = 0, 0.3, 0.7, 0.9$ and solve on the short rectangle

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq v \leq \frac{1}{2}, \quad (4.9)$$

and on the long rectangle

$$-1 \leq u \leq 0, \quad 0 \leq v \leq 1. \quad (4.10)$$

The coordinate refinements contain 17^2 , 33^2 , and 65^2 nodes. The Schwarzschild runs use 50 spherical points with retained degree five, while the finest Kerr results below use 86 spherical points with retained degree seven. Eight complete Picard sweeps are used in every case.

Because the exact metrics are known, no curvature differentiation is needed to define the primary error. At every angular and double-null grid point $q = (\theta, u_i, v_j)$, set

$$E_g(q) = \|\gamma_{\text{num}}(q) - \gamma_{\text{exact}}(q)\|_F, \quad (4.11)$$

where $\|\cdot\|_F$ is the Frobenius norm in the pole-free ambient representation of tangent two-tensors. Table 1 gives the median and arithmetic mean of E_g over all points of each 65×65 run. Schwarzschild and the zero-spin Kerr control agree with the explicit metric essentially to floating-point accuracy. For $a \neq 0$, the larger error is associated with the numerical angular reconstruction of the Kerr optical map; it remains of order 10^{-6} in the mean for all rotating cases considered here.

Table 1: Final section-metric error (4.11) on the finest coordinate grid.

Family	Domain	median E_g	mean E_g	max E_g
Schwarzschild	short	2.168×10^{-14}	2.387×10^{-14}	1.087×10^{-13}
Schwarzschild	long	3.782×10^{-13}	3.686×10^{-13}	9.207×10^{-13}
Kerr, $a = 0.0$	short	1.264×10^{-14}	1.313×10^{-14}	4.252×10^{-14}
Kerr, $a = 0.0$	long	3.445×10^{-13}	3.387×10^{-13}	7.693×10^{-13}
Kerr, $a = 0.3$	short	2.892×10^{-7}	5.727×10^{-7}	8.037×10^{-6}
Kerr, $a = 0.3$	long	1.120×10^{-6}	2.356×10^{-6}	3.918×10^{-5}
Kerr, $a = 0.7$	short	4.442×10^{-7}	9.360×10^{-7}	1.792×10^{-5}
Kerr, $a = 0.7$	long	1.888×10^{-6}	4.384×10^{-6}	1.030×10^{-4}
Kerr, $a = 0.9$	short	5.247×10^{-7}	9.864×10^{-7}	1.753×10^{-5}
Kerr, $a = 0.9$	long	2.446×10^{-6}	5.373×10^{-6}	1.022×10^{-4}

4.2 Schwarzschild horizon in static and Kruskal coordinates

The second experiment keeps the Schwarzschild geometry (4.1) fixed and changes only the coordinate system. It separates loss of numerical conditioning caused by a degenerating

chart from the regular geometry of the event horizon. In the static double-null chart (4.3), the additive constant in r_* is chosen so that

$$r = 2M(1 + \epsilon) \quad \text{at} \quad v - u = \frac{1}{2}. \quad (4.12)$$

The four offsets

$$\epsilon = 10^{-1}, 10^{-2}, 10^{-4}, 10^{-6} \quad (4.13)$$

therefore give

$$\min \Omega^2 = \frac{\epsilon}{1 + \epsilon}. \quad (4.14)$$

Although every one of these rectangles remains in the exterior $r > 2M$, the last case places the closest numerical sphere only $2M\epsilon = 2 \times 10^{-6}$ from the horizon when $M = 1$. The lapse consequently becomes small and the static coordinate representation becomes increasingly ill-conditioned.

For the horizon-crossing test, define shifted Kruskal coordinates

$$U = u + \frac{3}{4}, \quad V = v + 1. \quad (4.15)$$

The relation between r and (U, V) and the corresponding double-null metric are

$$UV = \left(1 - \frac{r}{2M}\right) e^{r/(2M)}, \quad \mathbf{g}_{\text{Sch}} = -\frac{32M^3}{r} e^{-r/(2M)} dU dV + r^2 \gamma, \quad (4.16)$$

or, in the notation of this paper,

$$\Omega^2 = \frac{8M^3}{r} e^{-r/(2M)}, \quad \gamma = r^2 \gamma, \quad b = 0. \quad (4.17)$$

Unlike the static lapse, this Ω is positive and regular at $r = 2M$. The principal Lambert branch gives the explicit inversion

$$r = 2M \left[1 + W_0 \left(-\frac{UV}{e} \right) \right]. \quad (4.18)$$

All five cases are solved on the same numerical rectangle

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq v \leq \frac{1}{2}, \quad (4.19)$$

using 17^2 , 33^2 , and 65^2 coordinate grids, 50 spherical points, retained angular degree five, and eight Picard sweeps. For the Kruskal case, (4.19) corresponds to

$$-\frac{1}{4} \leq U \leq \frac{1}{4}, \quad 1 \leq V \leq \frac{3}{2}. \quad (4.20)$$

It crosses the future event horizon $U = 0$ and covers $1.67549 \leq r \leq 2.24420$ for $M = 1$.

We again use the direct metric error E_g defined in (4.11). Table 2 reports its median and arithmetic mean on the finest grid. The static-coordinate metric itself is reproduced to roundoff even as Ω^2 approaches zero. The Kruskal calculation also remains highly accurate across the horizon, with a mean section-metric error of 1.12×10^{-12} . The supplementary curvature checks use first null derivatives of the weighted connection variables, with the metric-connection consistency reported separately. Derivative-based curvature diagnostics are more sensitive to the static lapse degeneration; this is a coordinate-conditioning effect and does not indicate a curvature singularity at $r = 2M$.

Table 2: Final section-metric error for the Schwarzschild horizon tests on the 65×65 coordinate grid.

Coordinates	median E_g	mean E_g	max E_g
Static, $\epsilon = 10^{-1}$	2.627×10^{-15}	2.800×10^{-15}	1.358×10^{-14}
Static, $\epsilon = 10^{-2}$	2.473×10^{-15}	2.749×10^{-15}	1.435×10^{-14}
Static, $\epsilon = 10^{-4}$	1.858×10^{-15}	1.960×10^{-15}	8.129×10^{-15}
Static, $\epsilon = 10^{-6}$	1.776×10^{-15}	1.873×10^{-15}	7.501×10^{-15}
Kruskal crossing	1.134×10^{-13}	1.122×10^{-12}	2.923×10^{-11}

4.3 Schwarzschild interior and approach to $r = 0$

The third experiment uses the Schwarzschild solution in the regular Kruskal form (4.16), now entirely inside the event horizon. We shift the Kruskal coordinates by

$$U = u + 2, \quad V = v + \frac{1}{5}, \quad (4.21)$$

and use the same relation

$$UV = \left(1 - \frac{r}{2M}\right) e^{r/(2M)}. \quad (4.22)$$

The singular boundary $r = 0$ corresponds to $UV = 1$. To approach it without including it, let

$$x_\epsilon = (1 - \epsilon)e^\epsilon, \quad v_0(u) = \frac{x_\epsilon}{u + 2} - \frac{1}{5}, \quad (4.23)$$

and introduce a fixed computational coordinate ξ through

$$v = \xi v_0(u). \quad (4.24)$$

The numerical domain is therefore the curved characteristic region

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq \xi \leq 1, \quad 0 \leq v \leq v_0(u). \quad (4.25)$$

On its future boundary $\xi = 1$ one has

$$UV = x_\epsilon, \quad r = 2M\epsilon. \quad (4.26)$$

We take

$$\epsilon = 2^{-1}, 2^{-2}, \dots, 2^{-6}. \quad (4.27)$$

For $M = 1$, the six future boundaries thus lie at $r = 1, 1/2, 1/4, 1/8, 1/16, 1/32$. The opposite characteristic face has $r \leq 1.84064$, so every case lies strictly within the Schwarzschild event horizon.

The map (4.24) is discretized with tensor-product Chebyshev–Lobatto grids containing 17^2 , 33^2 , and 49^2 points. The physical derivatives include both Jacobian terms generated by the u -dependence of v_0 . Each case receives at most 20 Picard sweeps. If an undamped update would make the radius nonpositive, dyadic backtracking chooses the largest positive, nonincreasing update. This stabilization prevents an algebraic breakdown, but it does not by itself establish convergence or accuracy near $r = 0$.

Since the solution is spherically symmetric,

$$\gamma_{\text{num}} = r_{\text{num}}^2 \overset{\circ}{\gamma}, \quad \gamma_{\text{exact}} = r_{\text{exact}}^2 \overset{\circ}{\gamma}. \quad (4.28)$$

Accordingly, the pointwise metric error (4.11) reduces to

$$E_g = \sqrt{2} \left| r_{\text{num}}^2 - r_{\text{exact}}^2 \right|, \quad (4.29)$$

Table 3: Final section-metric error in the Schwarzschild-interior experiment. Here $r_{\text{end}} = 2M\epsilon$ is the exact radius of the future boundary and $M = 1$.

ϵ	r_{end}	median E_g	mean E_g	max E_g
2^{-1}	1	2.026×10^{-7}	5.158×10^{-7}	2.208×10^{-6}
2^{-2}	1/2	2.810×10^{-5}	4.039×10^{-5}	1.364×10^{-4}
2^{-3}	1/4	2.212×10^{-3}	4.203×10^{-3}	1.835×10^{-2}
2^{-4}	1/8	6.969×10^{-3}	1.234×10^{-2}	6.119×10^{-2}
2^{-5}	1/16	2.020×10^{-2}	3.743×10^{-2}	2.851×10^{-1}
2^{-6}	1/32	3.419×10^{-2}	8.845×10^{-2}	4.180×10^{-1}

because $\|\overset{\circ}{\gamma}\|_{\text{F}} = \sqrt{2}$ in the pole-free ambient representation. Table 3 reports the statistics of (4.29) on the 49^2 grid.

The first two future boundaries retain small absolute metric error. Beginning at $\epsilon = 2^{-3}$, however, the iteration does not settle within 20 sweeps and the error grows rapidly under further approach to the singularity. In the closest case, the exact minimum radius is $1/32$, whereas the computed minimum is 0.08127 ; the maximum relative radius error is 16.43 , and the maximum relative section-metric error is 3.03×10^2 . That case also requires a dyadic relaxation factor as small as $1/32$. These results demonstrate the current method's loss of accuracy near $r = 0$; the stabilized finite output is not evidence of convergence at the Schwarzschild singularity.

4.4 Strong outgoing short pulse and zero control

The fourth experiment leaves the class of explicit solutions. At the corner we prescribe the round section metric, unit lapse, vanishing shift and torsion, and the Minkowski expansions,

$$\begin{aligned} \gamma(-1, 0) &= \overset{\circ}{\gamma}, & \Omega(-1, 0) &= 1, & b(-1, 0) &= 0, & \zeta(-1, 0) &= 0, \\ (\Omega \text{tr } \chi)(-1, 0) &= 2, & (\Omega \text{tr } \underline{\chi})(-1, 0) &= -2. \end{aligned} \quad (4.30)$$

On the outgoing initial hypersurface, a low-band trace-free tensor is formed from two polynomial hemisphere tensors X_+ and X_- and two disjoint profiles q_+ and q_- :

$$\Omega \hat{\chi}(-1, v) = \mathcal{T}_{q_+(v)X_+ + q_-(v)X_-}[\gamma(-1, v)]. \quad (4.31)$$

Here q_+ has the admissible square-root behavior at $v = 0$, q_- is a smooth interior bump, and both are calibrated on $0 \leq v \leq 1/2$ so that the hemisphere integral has strength one after division by 3.2. Only the leading part of this pulse is evolved, on the rectangle

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq v \leq 0.005. \quad (4.32)$$

The zero control uses the identical construction and discretization with $q_+ = q_- = 0$.

The perturbation in the strong run is not numerically negligible. Using the computed metric and area form, we measure

$$\|\hat{\chi}(-1, 0.005)\|_{L^2(S, \gamma)} = \left(\int_{S_{-1, 0.005}} |\Omega^{-1} \Omega \hat{\chi}|_{\gamma}^2 d\mu_{\gamma} \right)^{1/2} = 3.75480, \quad (4.33)$$

while the pointwise maximum is 1.26734 . Thus the good residual reported below is not a consequence of evolving an effectively zero datum.

We use 17 nodes in the logarithmic coordinate $\tau = -\log(-u)$ and 33 nodes in $s = \sqrt{2}v$, distributed over two and four degree-eight elements, respectively. The angular discretization has 550 sphere points, retained degree $L = 10$, work degree $W = 20$, and differentiation degree $D = 21$. Both runs complete six Picard sweeps. Since no explicit spacetime is available, the accuracy statistic is the $L_{u,v}^\infty L^2(S)$ norm of the independently differentiated first-order Ricci residual on the stencil-protected interior. The audit transfers scalar, vector, and tensor fields in their typed harmonic spaces through degree 20 onto 562 sphere points, and raises each coordinate element degree by three. The protected metric-connection component defects are 1.75×10^{-9} in the incoming direction and 2.77×10^{-7} in the outgoing direction.

Table 4: Iteration update and independent Ricci residual for the strong outgoing pulse and its zero control. The pointwise column is the maximum sphere-point residual on the same protected region.

Data	final update	$L_{u,v}^\infty L^2(S)$	protected pointwise maximum
Strong pulse	1.688×10^{-10}	2.617×10^{-5}	3.297×10^{-5}
Zero control	2.927×10^{-13}	1.973×10^{-9}	1.271×10^{-9}

Every boundary spectral-deferred-correction element is accepted. For the strong pulse, the maximum collocation and overgrid boundary defects are 6.36×10^{-16} and 1.47×10^{-8} , respectively. In the zero control, moving the residual mask farther from the short-pulse endpoint reduces the $L_{u,v}^\infty L^2(S)$ residual from 1.973×10^{-9} to 1.737×10^{-9} . The exact Minkowski control has the same numerical floor, 1.973×10^{-9} . The contrast between this numerical floor and the strong-pulse value isolates the error associated with evolving the nonspherical datum.

4.5 Crossed characteristic shears

The fifth experiment prescribes nonspherical shears on both initial null hypersurfaces. Let

$$A_1 = \text{diag}\left(-\frac{1}{2}, -\frac{1}{2}, 1\right), \quad A_2 = \text{diag}(1, -1, 0), \quad f_j(n) = n^\top A_j n, \quad (4.34)$$

and define the explicitly normalized round-sphere tensors

$$X_1 = \frac{\sqrt{2}}{3} \text{tf } \nabla^2 f_1, \quad X_2 = \frac{1}{2\sqrt{2}} \text{tf } \nabla^2 f_2. \quad (4.35)$$

If $\mathcal{T}_X[\gamma]$ denotes the symmetric transfer of X to γ followed by removal of its γ -trace, the prescribed data are

$$\Omega \widehat{\chi}(-1, v) = v^{1/2} \mathcal{T}_{X_1}[\gamma(-1, v)], \quad \Omega \widehat{\underline{\chi}}(u, 0) = (u+1)^{1/2} \mathcal{T}_{X_2}[\gamma(u, 0)]. \quad (4.36)$$

At the corner, $\gamma = \overset{\circ}{\gamma}$, $\Omega = 1$, $\Omega \text{tr } \chi = 2$, $\Omega \text{tr } \underline{\chi} = -2$, and $\zeta = 0$. The lapse is one on both initial faces and $b(u, 0) = 0.1 \partial_\varphi$; the remaining face data are obtained from the characteristic constraints.

Although the shears in (4.36) vanish continuously at the corner, the corresponding initial curvature is singular. The outgoing shear equation contains $\nabla_4 \widehat{\chi}$ and the incoming shear equation contains $\nabla_3 \widehat{\underline{\chi}}$. Their leading transverse derivatives satisfy

$$\nabla_4 \widehat{\chi} = O(v^{-1/2}), \quad \nabla_3 \widehat{\underline{\chi}} = O((u+1)^{-1/2}). \quad (4.37)$$

Consequently, both R_{4A4B} on $u = -1$ and R_{3A3B} on $v = 0$ are unbounded and hence infinite at the corner in the limiting sense. This is a singularity of the prescribed initial curvature, not a large-amplitude divergence of the shears themselves. Indeed, at the opposite endpoints of the two initial faces,

$$\|\widehat{\chi}(-1, \tfrac{1}{2})\|_{L^2(S, \gamma)} = 2.727749, \quad \|\widehat{\chi}(-\tfrac{1}{2}, 0)\|_{L^2(S, \gamma)} = 0.785641. \quad (4.38)$$

The solution region is the full rectangle

$$-1 \leq u \leq -\tfrac{1}{2}, \quad 0 \leq v \leq \tfrac{1}{2}. \quad (4.39)$$

We discretize the square-root coordinates $t = \sqrt{2(u+1)}$ and $s = \sqrt{2v}$ using six degree-ten elements in each direction, giving a 61×61 coordinate grid. The angular calculation uses 300 points with $L = 8$, $W = 14$, and $D = 15$. The evolution proceeds in six successive v -slabs. Their terminal weighted updates are

$$(1.850, 4.897, 8.694, 4.404, 6.984, 4.761) \times 10^{-9}, \quad (4.40)$$

all below the prescribed 10^{-8} tolerance.

Table 5: Aggregate Ricci residual for the crossed-shear experiment.

Region	Number of sections	median E_{Ric}	mean E_{Ric}	max E_{Ric}
Protected mask	144	3.960×10^{-4}	5.914×10^{-4}	3.432×10^{-3}
Open grid	3481	1.287×10^{-3}	8.947×10^0	1.371×10^3

Let $E_{\text{Ric}}(u, v)$ denote the sum of the sectionwise $L^2(S_{u,v})$ norms of the six weighted Ricci quantities listed in the curvature-residue subsection. Table 5 compares this aggregate residual on the protected mask with the same statistic on the open coordinate grid, excluding the four outer endpoint lines.

The protected mask is constructed separately in the two characteristic coordinates. In each coordinate it removes the complete first and last spectral elements and three LGL nodes on each side of every remaining element interface. The two-dimensional mask is the tensor product of these one-dimensional masks, leaving 12 retained nodes in each direction and hence 144 sections. Its purpose is to omit locations at which the independent derivative reconstruction encounters the square-root endpoint, a terminal one-sided boundary, or an element interface; the open-grid row retains the nearby endpoint layers and element interfaces, but excludes the endpoint lines themselves. The minimum section-metric eigenvalue is 2.938×10^{-2} , and all immutable characteristic traces are restored to round-off, with maximum incoming $\Omega \text{tr} \chi$ mismatch 1.78×10^{-15} . On the protected mask the metric-connection defects $\partial_u g + \mathcal{L}_b g - 2\Omega \chi$ and $\partial_v g - 2\Omega \chi$ have maximum absolute ambient components 3.10×10^{-4} and 1.85×10^{-5} , respectively.

Figure 1 includes the unprotected layers of the open grid. The extreme null-curvature components can diverge at the initial corner, but they are not differentiated to obtain this residual. The large unprotected values instead expose errors in the first-order differentiation near the fractional-power endpoints and element interfaces. They are not used as an interior accuracy statistic; the protected values are reported in Table 5.

5 Results for Einstein Scalar-field Equations

We test the Einstein-scalar-field iteration in three complementary regimes. Experiment 6 uses the explicit Fisher-Janis-Newman-Winicour family to measure the section-metric

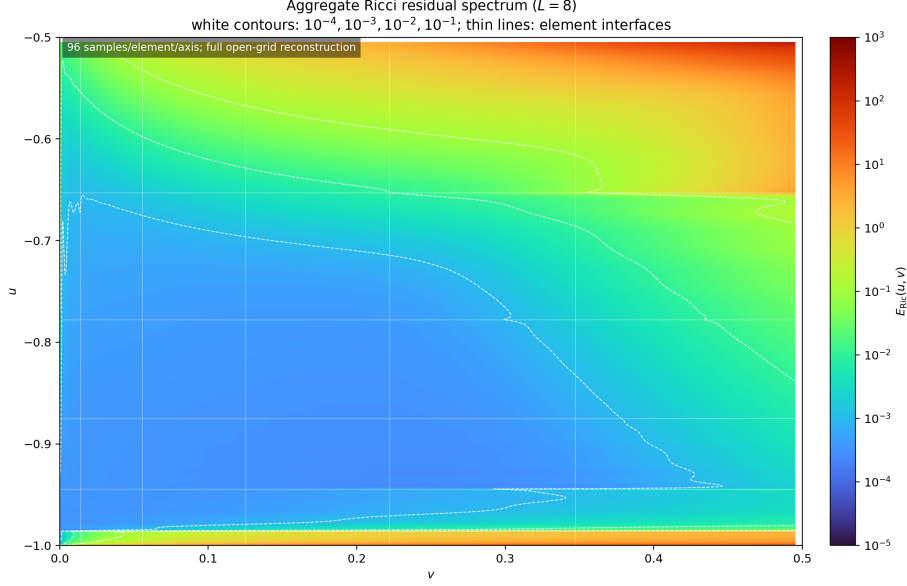


Figure 1: Spectrum of the aggregate Ricci residual $E_{\text{Ric}}(u, v)$. Colors show the elementwise spectral interpolation of $\log_{10} E_{\text{Ric}}$ on the open coordinate grid; white contours mark 10^{-4} , 10^{-3} , 10^{-2} , and 10^{-1} , and the thin lines are coordinate-element interfaces.

error directly. Experiment 7 replaces spherical symmetry by smooth angular lapse, shift, shear, and scalar data; without an explicit reference spacetime, its accuracy is measured by the independently differentiated first-order curvature residual $\text{Ric} - d\phi \otimes d\phi$. Experiment 8 strengthens the scalar pulse, covers a curved characteristic region by an overlapping inner atlas, locates a trapped region, and reconstructs the associated apparent horizon as a tube of marginally outer trapped surfaces.

5.1 Fisher–Janis–Newman–Winicour benchmarks

The Fisher–Janis–Newman–Winicour family is the standard static, spherically symmetric solution of the Einstein–massless-scalar equations. For $\sigma > 0$, $0 < \nu \leq 1$, and

$$f(r) = 1 - \frac{\sigma}{r}, \quad c_\nu = \sqrt{\frac{1 - \nu^2}{2}}, \quad (5.1)$$

the metric and scalar field in static coordinates are

$$\mathbf{g}_{\text{JNW}} = -f^\nu dt^2 + f^{-\nu} dr^2 + r^2 f^{1-\nu} \gamma^\circ, \quad (5.2)$$

$$\phi = c_\nu \log f. \quad (5.3)$$

They satisfy $\text{Ric}_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi$ with the normalization used here. Introduce an optical radius and null coordinates by

$$\frac{dr_*}{dr} = f^{-\nu}, \quad t = u + v, \quad r_* = v - u - \frac{1}{2}, \quad r(0) = 2\sigma. \quad (5.4)$$

Then (5.2) takes the double-null form

$$\mathbf{g}_{\text{JNW}} = -4\Omega^2 du dv + \gamma, \quad \Omega^2 = f^\nu, \quad \gamma = r^2 f^{1-\nu} \gamma^\circ, \quad b = 0. \quad (5.5)$$

We set $\sigma = 1$ and use

$$\nu = 0.99, 0.8, 0.5, 0.2 \quad (5.6)$$

on the numerical rectangle

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq v \leq \frac{1}{2}. \quad (5.7)$$

Thus $0 \leq r_* \leq 1$ and $2 \leq r \leq 2.89832$ across the four cases, so the domain remains strictly outside the curvature singularity $r = \sigma$. The characteristic traces are extracted from (5.5), while the interior is generated by eight Picard sweeps. Coordinate refinements use two, four, and eight degree-six elements in each null direction; the finest grid has 49×49 coordinate nodes. A separate $\nu = 1$ control reduces to Schwarzschild with mass $M = \sigma/2 = 1/2$ and has identically vanishing scalar variables.

As in the vacuum benchmarks, define the pointwise section-metric error by

$$E_g(q) = \|\gamma_{\text{num}}(q) - \gamma_{\text{exact}}(q)\|_{\text{F}}. \quad (5.8)$$

Table 6 gives its statistics over all sphere and coordinate nodes of the finest grid.

Table 6: Final section-metric error for the Fisher–JNW benchmarks on the 49×49 coordinate grid.

ν	median E_g	mean E_g	max E_g
0.99	7.211×10^{-13}	7.372×10^{-13}	1.717×10^{-12}
0.80	1.856×10^{-12}	1.740×10^{-12}	3.751×10^{-12}
0.50	7.676×10^{-12}	7.475×10^{-12}	2.790×10^{-11}
0.20	3.576×10^{-11}	4.513×10^{-11}	5.641×10^{-10}

The error increases as ν moves away from the Schwarzschild limit and the scalar field becomes stronger, but even the $\nu = 0.2$ case retains a maximum absolute metric error below 5.7×10^{-10} .

5.2 Nonspherical scalar characteristic data

The seventh experiment prescribes nonspherical characteristic data on

$$-1 \leq u \leq -\frac{1}{2}, \quad 0 \leq v \leq 0.04. \quad (5.9)$$

Let $r = -u$ on the incoming initial hypersurface, and let (x, y, z) be Cartesian coordinates restricted to the unit sphere. Define the smooth fields

$$Y = \frac{3z^2 - 1}{2}, \quad V = \nabla_{\hat{\gamma}}(xz), \quad Q = (\nabla_{\hat{\gamma}}^2(x^2 - y^2))^{\text{tf}}, \quad P_0 = \frac{8\sqrt{2}}{5}. \quad (5.10)$$

The free incoming data are

$$\gamma(u, 0) = r^2 \hat{\gamma}, \quad \Omega^2(u, 0) = r^{1/4}(1 + 0.01Y), \quad b(u, 0) = 0.05V. \quad (5.11)$$

On the outgoing initial hypersurface we prescribe

$$\Omega^2(-1, v) = 1 + 0.01Y, \quad (5.12)$$

$$(\Omega e_4 \phi)(-1, v) = P_0 + v^{0.1}, \quad (5.13)$$

$$\Omega \hat{\chi}(-1, v) = 0.1v^{0.1} \mathcal{T}_Q[\gamma(-1, v)]. \quad (5.14)$$

The normalizations are fixed analytically, independently of the sphere grid. The free metric, log-lapse, shift, and tensor profiles are represented in the prescribed angular Galerkin

spaces before completing the face constraints. At the corner $\phi = 0$, $\Omega \operatorname{tr} \chi = 1.6$, $\zeta = 0$, and $\Omega\omega = 0$. The positive branch of $\Omega e_3 \phi$ and all remaining face fields are determined by the Einstein–scalar characteristic constraints. The functions in (5.11)–(5.12) are globally smooth in angle, while the power coordinate

$$\tau = -\log(-u), \quad s = (v/0.04)^{0.1} \quad (5.15)$$

resolves their prescribed $v^{0.1}$ corner behavior.

There is no explicit spacetime for comparison. After independent angular resampling and coordinate overgrid refinement we therefore use

$$E_{\text{Ric}}(u, v) = \|\text{Ric} - d\phi \otimes d\phi\|_{L^2(S_{u,v})}. \quad (5.16)$$

Here the Ricci components are computed from first null derivatives of the resampled weighted connection variables, with metric–connection and scalar consistency and the scalar wave residual checked separately. The positive norm squares the null components with weights 1, 1, 2, 1, 1 for 33, 44, 34, 3A, 4A, respectively, and uses the full tensor norm for AB ; it is not a Lorentzian contraction. For coordinate comparison, the audit uses the union of the two- and three-element source partitions, with degrees 11 and 14 in τ and s . All three solutions are therefore evaluated on the same 45×57 grid. The protected region removes three nodes from each outer endpoint and each audit-element interface, requires $s \geq 0.6$, and contains 198 sections. The scalar, vector, and tensor fields are transferred in their typed spaces; for the central angular discretization the transfer degree is 16 on 362 independent sphere points. Table 7 shows coordinate convergence with the central $L = 8$, $W = 16$, 350-point angular discretization held fixed.

Table 7: Protected independent curvature residual for Experiment 7. Grid sizes refer to the evolved state before independent overgrid resampling.

Coordinate grid	median E_{Ric}	mean E_{Ric}	max E_{Ric}
13×19	5.947×10^{-3}	9.907×10^{-3}	6.251×10^{-2}
17×23	2.677×10^{-3}	3.706×10^{-3}	1.361×10^{-2}
25×34	1.550×10^{-3}	2.676×10^{-3}	1.070×10^{-2}

All three coordinate runs complete six Picard sweeps, and the protected maximum decreases from 6.25×10^{-2} to 1.07×10^{-2} . At fixed central coordinate resolution, the sum of the protected Einstein and scalar-wave residual maxima decreases from 3.99×10^{-2} through 1.55×10^{-2} to 9.79×10^{-3} under angular refinement. Figure 2 displays the finest-coordinate audit. The plot includes the full open overgrid to expose the unresolved $v^{0.1}$ endpoint layer, whereas the statistics in Table 7 use only the protected region.

5.3 Scalar-pulse formation of a trapped region and apparent horizon

Experiment 8 keeps the incoming spherical geometry

$$\gamma(u, 0) = (-u)^2 \gamma^\circ, \quad \Omega^2(u, 0) = (-u)^{1/4}, \quad b(u, 0) = 0, \quad (5.17)$$

but selects the negative branch of the incoming scalar constraint. At the corner,

$$\phi = 0, \quad \Omega \operatorname{tr} \chi = 1.6, \quad \Omega \operatorname{tr} \underline{\chi} = -2, \quad (\Omega e_4 \phi)(-1, 0) = -2.2627416998. \quad (5.18)$$

Use the axisymmetric tensor $Q_{\text{ax}} = -(\nabla_\gamma^2 z^2)^{\text{tf}}$ and set

$$F(v) = \left(\frac{v}{v + 0.01} \right)^{0.1}, \quad (5.19)$$

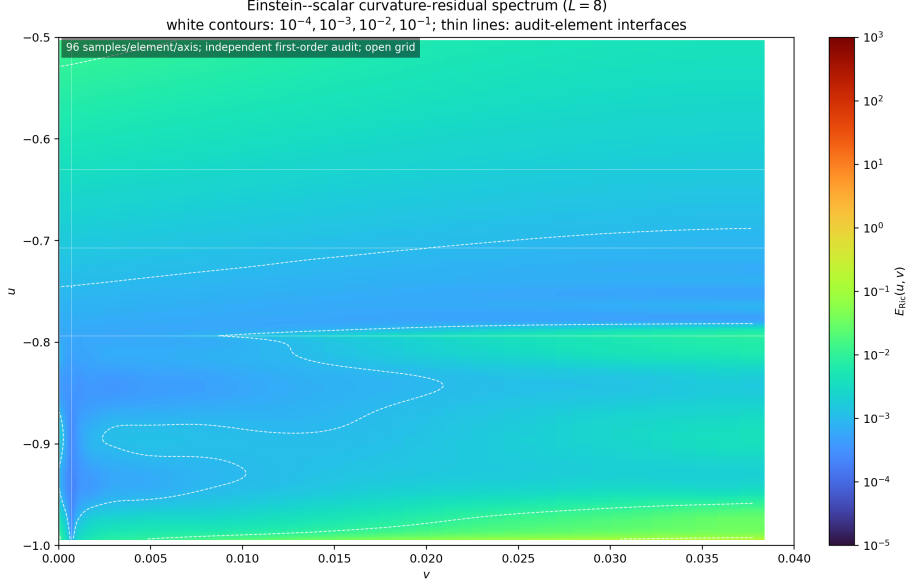


Figure 2: Spectrum of the independent Einstein-scalar curvature residual $E_{\text{Ric}}(u, v)$ for the finest coordinate run. Colors show elementwise spectral interpolation of $\log_{10} E_{\text{Ric}}$; white contours mark 10^{-4} , 10^{-3} , 10^{-2} , and 10^{-1} , and the thin lines mark audit-element interfaces.

the outgoing scalar and shear perturbations are

$$e_4\phi(-1, v) - e_4\phi(-1, 0) = -2F(v), \quad (5.20)$$

$$\Omega\hat{\chi}(-1, v) = 0.1F(v)\mathcal{T}_{Q_{\text{ax}}}[\gamma(-1, v)]. \quad (5.21)$$

Here $\Omega(-1, v) = 1$, so the first line is also the corresponding statement for the stored variable $\Omega e_4\phi$. The characteristic constraints determine the remaining data. On the initial faces the outgoing expansion is still positive, with minimum 0.880366, while $\text{tr } \underline{\chi}$ is negative. Thus the initial data themselves contain no trapped section.

The evolution region is changed to

$$\mathcal{D} = \left\{ (u, v) : -1 \leq u \leq -0.05, \quad 0 \leq v \leq \min \left(0.1(-u)^{25/24}, 0.05 \right) \right\}. \quad (5.22)$$

This cap follows the focusing scale and avoids evolving the excluded part of a bounding rectangle through the double-null caustic. The numerical solver remains rectangular: seven nested characteristic rectangles, each wholly contained in $\mathcal{D}$, form an inner staircase atlas covering 94.6694% of its coordinate area. Adjacent patches are compared in their overlaps; the largest absolute discrepancy among the checked primitive fields is 5.74×10^{-7} , while the section-metric discrepancy is below 1.53×10^{-8} .

Table 8 applies (5.16) on the same $s \geq 0.6$ protected region used above. These rows compare different atlas patches; the separate angular control tests the coordinate-refined anchor near $v = 0.04$.

Here u_{R} and v_{cap} are the right and upper boundaries of a patch. The protected mask requires $s \geq 0.6$ and removes three overgrid nodes at outer boundaries and at every source-element interface. The table therefore measures the independently reconstructed Einstein residual away from both the fractional-power corner layer and spectral differentiation interfaces.

Finally, we resample both null expansions independently on 1000 sphere points and search for a coordinate section satisfying

$$\sup_{S_{u,v}} \text{tr } \chi < 0, \quad \sup_{S_{u,v}} \text{tr } \underline{\chi} < 0. \quad (5.23)$$

Table 8: Protected independent curvature residual on the seven production patches of the Experiment 8 inner atlas.

u_R	v_{cap}	median E_{Ric}	mean E_{Ric}	max E_{Ric}
-0.52	0.050000	2.365×10^{-4}	2.097×10^{-3}	3.439×10^{-2}
-0.45	0.043527	2.141×10^{-4}	1.605×10^{-3}	1.915×10^{-2}
-0.35	0.033502	1.418×10^{-4}	1.096×10^{-3}	1.268×10^{-2}
-0.25	0.023597	8.421×10^{-5}	7.923×10^{-4}	6.761×10^{-3}
-0.15	0.013860	5.655×10^{-5}	6.318×10^{-4}	4.317×10^{-3}
-0.10	0.009085	3.761×10^{-5}	5.097×10^{-4}	4.923×10^{-3}
-0.05	0.004413	9.847×10^{-6}	1.451×10^{-4}	2.446×10^{-3}

The atlas contains such sections across several patches, producing a sampled trapped region rather than a single endpoint candidate. Direct four-metric evaluation at the reference section

$$(u, v) = (-0.4696271025, 0.04), \quad (5.24)$$

gives

$$\sup_{S_{u,v}} \theta_+ = -0.0802152, \quad \sup_{S_{u,v}} \theta_- = -4.28011. \quad (5.25)$$

Both null expansions are therefore strictly negative on this entire two-sphere. The independent $L = 7$, $W = 14$, 300-point angular control gives -0.0802016 and -4.27999 , respectively.

To reconstruct the boundary of the trapped region, on each incoming cone $v = v_0$ we solve the angular graph equation

$$\theta_+[u = h(\theta^A), v_0] = 0. \quad (5.26)$$

The expansion is calculated from the four-metric connection and graph embedding and does not reuse the evolved expansion variable. A section is retained only if the nonlinear solve succeeds within the patch interior, $\|\theta_+\|_{L^\infty} \leq 10^{-5}$, θ_- is strictly negative, and small inward and outward constant displacements bracket θ_+ . The search starts from the outermost detected constant-section bracket; this does not establish outermostness among arbitrary angular graphs.

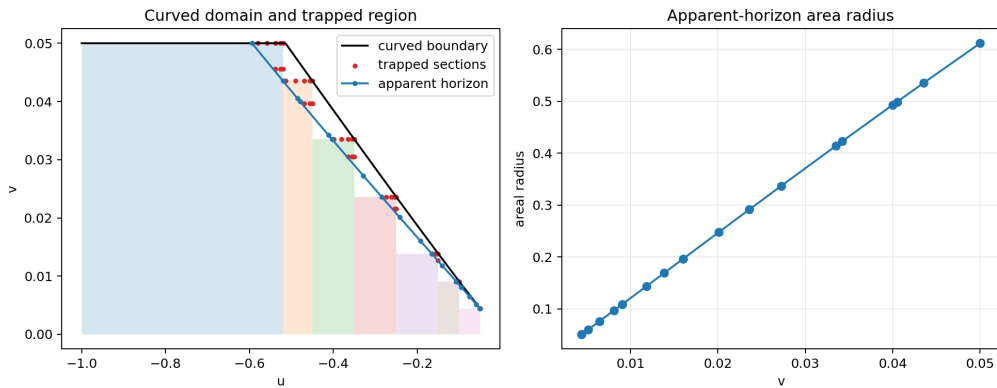


Figure 3: Experiment 8 curved characteristic region, inner rectangular atlas, sampled trapped sections, and reconstructed apparent horizon. Shading shows the atlas union, red points are trapped sections, and the blue curve joins the reconstructed MOTSs. The right panel shows their areal radii.

The resulting apparent horizon contains 18 verified MOTSs spanning $0.00441 \leq v \leq 0.05$. The mean graph location runs from $u = -0.0504104$ to $u = -0.5938472$, while the

areal radius increases from 0.0512842 to 0.6118550. Over the complete tube,

$$\max_v \|\theta_+\|_{L^\infty(S)} = 8.47 \times 10^{-6}, \quad \max_v \sup_{S_{h(v),v}} \theta_- = -3.38460, \quad (5.27)$$

and the maximum peak-to-peak angular deformation of h is 1.33×10^{-4} . At $v = 0.04$, the degree-five coordinate-refined MOTS has

$$\bar{h} = -0.479401999, \quad |S| = 3.05516039, \quad r_S = 0.493073969. \quad (5.28)$$

Refining the characteristic grid changes h by 1.50×10^{-7} in maximum norm and changes the area by 6.54×10^{-7} relatively. The 300-point angular control changes the mean location by 7.80×10^{-10} and the area by 3.29×10^{-9} relatively. A degree-three horizon trace agrees in mean location to 2.97×10^{-8} but leaves $\|\theta_+\|_{L^\infty}$ as large as 9.40×10^{-3} ; degree four is therefore required to resolve the nonspherical correction.

6 Conclusion and Outlook

We have developed a first-order Picard iteration for the characteristic initial value problem for the Einstein vacuum and Einstein–massless-scalar equations in double-null gauge. The construction follows the geometric dependency structure of the equations and combines characteristic constraint solves with spectral elements in the two null coordinates, pole-free angular operators, and independent first-order residual audits with separate metric–connection consistency checks. This provides a unified numerical framework for exact benchmarks and for characteristic data with no explicit interior solution.

The regular exact tests cover vacuum Schwarzschild and Kerr geometries and the scalar Fisher–JNW family. Metric errors remain small in regular domains and across a Schwarzschild horizon. The Schwarzschild-interior experiment also identifies the present method’s loss of accuracy as the curvature singularity is approached. Beyond the explicit families, the vacuum calculations evolve a strong nonspherical outgoing perturbation and crossed characteristic data with singular limiting initial curvature. The Einstein–scalar calculations further demonstrate coordinate convergence for nonspherical, low-regularity characteristic data on the protected numerical region.

The final experiment gives numerical evidence for a trapped region and reconstructs its apparent horizon. On the curved characteristic domain, both angular discretizations identify the section

$$(u, v) = (-0.4696271025, 0.04), \quad (6.1)$$

where the two future null expansions are strictly negative on the entire sphere. Eighteen independently reconstructed MOTSs form a sampled apparent-horizon tube over $0.00441 \leq v \leq 0.05$. Coordinate refinement changes its $v = 0.04$ graph by 1.50×10^{-7} in maximum norm, while an independent higher-angular-band evolution changes its mean location by 7.80×10^{-10} . The distinction between exact metric errors, independently evaluated curvature residuals, trapped-section sign tests, and the graph MOTS equation remains essential when interpreting the experiments.

Three directions are particularly important for future work. First, the angular MOTS solve should be extended with adaptive coordinate refinement toward the singular end-point and across alternative foliations, thereby resolving the anisotropic apparent horizon more completely. Second, varying the scalar-pulse and anisotropic-perturbation amplitudes would reveal the threshold for trapping and the dependence of the first trapped location on the initial data. Third, improved coordinates, adaptive refinement, and more robust iteration strategies are needed to improve the approximation near curvature singularities. These developments would extend the present double-null framework from a local numerical construction toward a more precise study of anisotropic black-hole formation and singular spacetime geometry.

Acknowledgments

SW is supported by the NUS President Graduate Fellowship.

Code availability

The source code, experiment configurations, and instructions for reproducing the numerical experiments are available at

<https://github.com/Shengrong-Wu/Numerical-Einstein-Equations>.

The numerical implementation is recorded at commit `f158a4e`. The generated tables and figures include provenance records identifying their evolution and diagnostic software versions and source-artifact hashes.

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